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# A Unified Convergence Theory for Non-Stationary Reinforcement Learning

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## Abstract

1 Non-stationary reinforcement learning lacks a unified convergence theory: variation-  
2 budget regret bounds, sliding-window methods, tempo-and-forgetting analyses,  
3 two-term decompositions, and causal-RL frameworks each address fragments with-  
4 out composing. We present a *compatible bundle of guarantees* that delivers all  
5 three properties — non-stationarity persistence, explicit metric structure on policy  
6 space, and on-policy learnability — *jointly*. Each of four components supplies  
7 a distinct epistemic property: (i) a two-gap diagnostic separating goal-feasibility  
8 from policy-quality routes corrective action; (ii) a point-mass reverse-KL/TV iden-  
9 tity under deterministic optimum,  $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \parallel Q)}$  — an *exact*  
10 algebraic identity that lies strictly below the Pinsker and Bretagnolle–Huber en-  
11 velopes at this corner — supplies a coordinate-optimal regret coordinate; (iii) a  
12 multi-factor strategic tempo  $\bar{\mathcal{T}}_\Sigma = \min_{(i,j)} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$  with structural sur-  
13 vival inequality  $\mathcal{T}_\Sigma > \rho_\Sigma / R_\Sigma$  supplies persistence; (iv) closed-loop interventional  
14 access under sequential ignorability supplies on-policy learnability via the Pearl  
15 causal hierarchy. Composition yields a Best-of-Both-Worlds dynamic regret theo-  
16 rem  $\tilde{O}(\min\{V_{\max} N_h \sqrt{(B_T + 1)T}, V_{\max} N_h (V_T + 1)^{1/3} T^{2/3}\})$  — automatically  
17 adapting between piecewise-stationary segment count  $B_T$  and continuous variation  
18 budget  $V_T$  without prior knowledge of either; the  $V_T$  exponent is information-  
19 theoretically near-optimal at the deterministic- $\pi^*$  corner. A structural failure-  
20 class theorem identifies the *gain-decay* updates (count-accumulating Bayesian  
21 schemes, Robbins–Monro gradient methods) as universally violating the forgetting  
22 prerequisite for any positive disturbance rate; finite-gain mechanisms (constant-  
23 step stochastic approximation, sliding windows, bounded memory, block restart)  
24 face *bidirectional* thresholds instead. The deterministic-optimum scope extends  
25 perturbatively to  $\epsilon$ -stochastic and softmax-regularized policies with  $O(\epsilon \log(1/\epsilon))$   
26 and  $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$  corrections (exponentially small with polynomial pref-  
27 actor). Theory only; we make the ProST reduction rigorous at the steady-state  
28 sector-parameter level, lifting it from a single-factor hyperparameter claim to a  
29 multi-factor structural-threshold claim.

## 1 Introduction

31 Non-stationary reinforcement learning has matured along four largely separate research tracks. Each  
32 has produced rigorous results; none has been composed with the others into a single convergence  
33 theory.

34 The four tracks are visible in the recent literature as parallel lineages. **Variation-budget dynamic**  
35 **regret under drift**<sup>[1–4]</sup> gives sublinear dynamic regret under bounded total variation in reward  
36 and transition dynamics. **Two-term regret decompositions**<sup>[5–7]</sup> split the dynamic regret along an

37 *exploration-vs-adaptation* axis, isolating the cost of confidence-set construction from the cost of  
 38 tracking a moving optimum. **Tempo and forgetting analyses**<sup>[8–12]</sup> make policy-update timing or  
 39 evidence-discount rate an explicit convergence variable. **Causal and interventional access**<sup>[13–18]</sup>  
 40 uses causal-graph structure to sharpen sample complexity, but operates in stationary settings only.

41 The information-theoretic regret literature<sup>[19–24]</sup> traverses these tracks at a different layer (entropy / mu-  
 42 tual information / information ratio / Pinsker / Hellinger) — but the *exact* reverse-KL / total-variation  
 43 identity at the deterministic-optimum corner, strictly improving both Pinsker and the Bretagnolle–Hu-  
 44 ber inequality<sup>[25]</sup> at that corner, is absent from the retrieved corpus (Appendix F).

45 These four tracks share an evident common ancestor — the dynamic-regret analysis of online MDPs  
 46 — but no published framework composes them along the *goal-feasibility-vs-policy-quality* axis with  
 47 an exact point-mass identity, structural survival inequality, and closed-loop causal access. The  
 48 fundamental question remains open:

49 *Can we design a unified convergence theory for non-stationary reinforcement*  
 50 *learning that handles non-stationarity, has explicit metric structure on policy space,*  
 51 *and is learnable from on-policy data?*

52 We present here a composition that delivers all three properties jointly. Composing four structural  
 53 elements — a two-gap diagnostic, a point-mass reverse-KL/TV identity, a multi-factor strategic tempo  
 54 with forgetting prerequisite, and closed-loop interventional access — yields a cumulative dynamic  
 55 regret theorem with rate  $\text{DynReg}(T) \leq \tilde{O}(N_h \sqrt{(B_T + 1)T})$  in the piecewise-stationary regime,  
 56 where  $B_T$  counts piecewise-stationary segment boundaries (kernel-and-reward changes) along the  
 57 realized trajectory and  $N_h$  is the planning horizon. The per-round coordinate of this rate is sharper  
 58 than both Pinsker and Bretagnolle–Huber at the deterministic- $\pi^*$  corner. The technical anchor is an  
 59 exact algebraic identity:  $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \parallel Q)}$ , which holds for any policy  $Q$  with  
 60  $Q(a^*) > 0$  and lies *strictly below* both the Pinsker envelope  $\sqrt{D_{\text{KL}}/2}$  and the Bretagnolle–Huber  
 61 envelope  $\sqrt{1 - e^{-D_{\text{KL}}}}$  at this corner.

## 62 1.1 Contribution

63 Each of four components supplies a *distinct epistemic property* the others cannot deliver; together  
 64 they form a *compatible bundle of guarantees*.

65 **Component 1 — corrective-action routing** The *two-gap diagnostic* splits the regret signal along the  
 66 *goal-feasibility-vs-policy-quality* axis — orthogonal to the exploration-vs-adaptation decompositions  
 67 of<sup>[5–7]</sup> — routing four regimes (success, strategy problem, capability limit, both) to revise-policy /  
 68 revise-model-class-or-horizon / revise-objective (Section 4).

69 **Component 2 — metric layer on policy space** Under deterministic optimum  $\pi^* = \delta_{a^*}$ , the per-  
 70 round regret coordinate is *exact*:  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \parallel Q)}$ , *coordinate-optimal among bounds*  
 71 *depending only on TV* and strictly below both Pinsker and Bretagnolle–Huber envelopes at this corner  
 72 (Appendix E). Deterministic scope extends perturbatively to  $\epsilon$ -greedy and softmax-regularized optima  
 73 with quantified corrections (Section 5.1).

74 **Component 3 — non-stationarity persistence** A *multi-factor strategic tempo* composed per  
 75 element of observation rate  $\nu$ , identifiability  $\iota$ , and update gain  $1 - \lambda$  yields a survival inequality  $\mathcal{T}_\Sigma >$   
 76  $\rho_\Sigma / R_\Sigma$  — below threshold, no schedule preserves the agent’s strategic substate against disturbance  
 77 (Section 5.2). A structural-class theorem identifies *gain-decay* updates (count-accumulating Bayesian,  
 78 vanishing-step Robbins–Monro) as universally violating the threshold; finite-gain mechanisms face  
 79 bidirectional ceilings (Appendix C.6). *Lifts* Lee et al. 2023<sup>[8]</sup>’s tempo result from hyperparameter to  
 80 structural-threshold claim.

81 **Component 4 — on-policy learnability** Under (C1) positivity, (C2) sequential ignorability ( $H_t$   
 82 d-separates  $a_t$  from  $o_{t+1}$  in  $G_{a_t}^-$ ), and (C3) known action mechanism, the closed-loop generates Pearl  
 83 Level-2 data by construction<sup>[26,27]</sup>, making Component 2’s regret bound *learnable on-policy* under  
 84 sufficient identifiability (Section 5.3). Three regimes A/B/C partition usable strength; coupled-goal  
 85 architectures violate (C2) (Section 6).

86 **Composition** Via block-Cauchy–Schwarz in piecewise-stationary case and Wei and Luo 2021<sup>[2]</sup>  
87 MASTER wrapping in continuous variation, the four components compose into a Best-of-Both-Worlds  
88 theorem  $\mathbb{E}[\text{DynReg}(T)] \leq \tilde{O}\left(V_{\max}N_h \cdot \min\left\{\sqrt{(B_T+1)T}, (V_T+1)^{1/3}T^{2/3}\right\}\right) + V_{\max}N_h(1 -$   
89  $p_{\text{id}})T$ , *automatically adapting* to  $B_T$  or  $V_T$  without prior knowledge; matching Mao et al. 2021<sup>[3]</sup>’s  
90 near-optimal  $V_T$  rate, with Besbes et al. 2014<sup>[28]</sup>’s lower bound applying at the deterministic- $\pi^*$   
91 corner. The bias term vanishes in Regime A.

## 92 1.2 Scope and limitations

93 **What “convergence” means here** We do not claim pointwise policy convergence — under genuine  
94 non-stationarity the optimum is itself moving. By *convergence* we mean: per-round regret-coordinate  
95 convergence within stationary blocks; cumulative dynamic regret with sublinear / vanishing-average  
96 rate when  $B_T = o(T)$  (Cesàro tracking  $\frac{1}{T} \sum_t (V_t^* - V(\pi_t)) \rightarrow 0$ ); and ultimate boundedness of  
97 the strategic mismatch  $\|\delta_\Sigma\|$  under the tempo threshold. Pointwise  $V(\pi_t) \rightarrow V^*$  is structurally  
98 unavailable under genuine non-stationarity; the Cesàro tracking statement is the right replacement.

99 **Theory only** No experiments. The point-mass identity is mathematically airtight (a two-line direct  
100 calculation that specializes the BH family at its deterministic- $\pi^*$  corner). The empirical grounding  
101 for the strategic-tempo claim is provided indirectly through reduction to<sup>[8]</sup> ProST as a special case  
102 (Section 5); at the steady-state sector-parameter level the reduction is rigorous.

103 **Canonical scope** The composition theorem holds under deterministic optimum policy, bounded  
104 value range, isolated optimum (so  $\Delta_{\min} > 0$ ), and a singular causal trajectory in the sense made  
105 precise in Section 3. The deterministic- $\pi^*$  scope extends *perturbatively* to  $\epsilon$ -stochastic and softmax-  
106 regularized optima (Section 5.1); tied-optimum is a degraded one-sided form (Appendix C.5). Each  
107 axis of the canonical scope degrades cleanly outside it (we discuss the failure modes in Section 6).

108 **Comparator regime** The cumulative rate covers both the piecewise-stationary  $B_T$  family and the  
109 continuous-variation  $V_T$  family of<sup>[3]</sup> via Best-of-Both-Worlds wrapping (Section 6).

## 110 2 Related Work

111 The four tracks named in Section 1 organize the literature.

112 **Variation-budget dynamic regret** <sup>[1–4,29]</sup> establish dynamic regret bounds under a *variation budget*  
113  $V_T$  on cumulative reward and transition variation.<sup>[3]</sup> achieves the near-optimal continuous-variation  
114 rate  $\tilde{O}(SA V_T^{1/3} H T^{2/3})$  via RestartQ-UCB;<sup>[1]</sup> treats the piecewise-stationary case with restart-on-  
115 change. Our cumulative rate  $\tilde{O}(N_h \sqrt{(B_T+1)T})$  sits in the piecewise-stationary family — same  
116 block-shape as Cheung-style restart-on-change — recovered as a corollary of composition with the  
117 per-round identity. The novelty is the *threshold form* of the strategic-tempo inequality (environment  
118 regimes where no  $\lambda \in (0, 1)$  stabilizes the sector-Lyapunov mismatch), which dynamic-regret-  
119 optimization analyses do not surface.

120 **Two-term regret decompositions along the exploration-vs-adaptation axis** <sup>[5–7]</sup> decompose the  
121 dynamic regret into a *confidence-set construction* term and an *adaptation-under-non-stationarity*  
122 term.<sup>[30]</sup> treats a structurally adjacent question — constraint-region feasibility — using a two-term  
123 decomposition along that axis. Same shape, different axis: ours is goal-feasibility-vs-policy-quality  
124 (the satisfaction gap distinguishes “the goal is unattainable from  $M_t$ ” from “the current policy is  
125 suboptimal”); theirs is uncertainty-source. Neither subsumes the other: a Long-Fei Li-style exploration  
126 term can be present in any of our  $2 \times 2$  cells, and an adaptation term likewise. The two axes are  
127 orthogonal.

128 **Tempo and forgetting** <sup>[8]</sup> (ProST) defines an agent tempo as the schedule of policy update times  
129  $\{t_1, \dots, t_K\}$  and computes the schedule that minimizes a dynamic regret upper bound.<sup>[9]</sup> (Pausing  
130 Policy Learning) refines this with non-zero hold durations.<sup>[10–12]</sup> establish forgetting / sliding-window  
131 analyses for non-stationary linear MDPs and bandits. We *lift* the ProST move along two axes  
132 simultaneously: single-factor  $\rightarrow$  multi-factor (per-element  $\nu \cdot \iota \cdot (1 - \lambda)$  rather than scalar update  
133 frequency); and hyperparameter-optimization  $\rightarrow$  structural-survival inequality (the threshold below

which no schedule persists, irrespective of which schedule is chosen). ProST recovers as the Beta-Bernoulli special case  $|E| = 1$ ,  $\nu = \iota = 1$ .

**Causal and interventional access** <sup>[13–18,31]</sup> use causal-graph structure to sharpen sample complexity in RL. Their setting is *stationary*. We compose the closed-loop interventional-access argument with non-stationarity. The connection between Pearl’s causal hierarchy<sup>[26]</sup> and the data-generating process of an agent in a feedback loop<sup>[27]</sup> is the technical bridge.

**Information-theoretic regret (cross-cutting)** <sup>[19–24,32]</sup> traverse the above tracks at a different layer, using entropy / mutual information / information ratio / Pinsker / Hellinger as the divergence backbone. The Russo–Van Roy information ratio uses Pinsker as its Cauchy–Schwarz follow-up; the IDS line and its descendants use mutual-information bounds. The Bretagnolle–Huber inequality<sup>[25]</sup> — well-known in lower-bound proofs via Le Cam’s method<sup>[33,34]</sup> — does not appear as an *upper-bound regret coordinate* in this corpus, and its exact point-mass corner (our Lemma 5.1) is *a fortiori* absent. We document the retrieval scope and methodology in Appendix F.

**Adjacent (satisficing / feasibility)** <sup>[35,36]</sup> use *satisficing* to mean “any policy above acceptance level  $\beta$  is acceptable.” Our  $\delta_{\text{sat}}$  is structurally distinct:  $V_O^{\min}$  is a property of the objective (set by the application domain), and  $\delta_{\text{sat}} > 0$  signals goal-relative *infeasibility* (no policy in  $\Pi$  meets the threshold), not policy-relative satisficing.

**Contemporaneous (post-March 2026)** <sup>[36,37]</sup>; cited and distinguished without empirical-comparison demand per the contemporaneous-work cutoff convention.

### 3 Preliminaries

We adopt strict minimalism: only the notation and assumptions used directly in the main result and the mechanism statements appear here. Convention-hierarchy details (one-step vs. receding-horizon vs. Bellman continuation), the action-gap matching argument, tied-optimum extensions, and the deterministic-trajectory deployment modes are deferred to Appendix C.

**Episodic non-stationary MDP** A finite-horizon non-stationary MDP  $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$  with state space  $\mathcal{S}$ , finite action space  $\mathcal{A}$ , time-indexed transition kernels  $P_t(\cdot \mid s, a)$  and reward functions  $r_t(s, a)$  allowed to vary across rounds  $t$ , and finite planning horizon  $N_h$ . Standard non-stationary-RL boundedness assumptions<sup>[11]</sup>: per-step reward in  $[0, 1]$ .

**Variation regimes** A variation budget  $V_T$  measures continuous variation in transitions and rewards. We work in the *piecewise-stationary* specialization: time partitions into  $B_T + 1$  maximal stationary intervals on which  $(P_t, r_t)$  is constant, with  $B_T := |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$  counting *kernel-stationary-segment boundaries* (the right counter for the base learner’s per-block stationary-regret guarantee in (A5); see Appendix A.6). The two variation regimes ( $B_T$  vs. continuous  $V_T$ ) coincide up to constants under abrupt-magnitude- $\delta$  piecewise-stationarity ( $V_T \asymp \delta B_T$ ); cumulative results carry through to continuous  $V_T$  via Wei-Luo MASTER black-box wrapping (Section 6).

**Policy distribution** The agent’s policy at round  $t$  is a distribution  $Q_t(\cdot \mid s)$  over actions; we write  $Q$  when state and round are understood. We adopt the canonical scope: the optimum policy at  $(M_t, s)$  is *deterministic*,  $\pi^*(\cdot \mid M_t, s) = \delta_{a^*(s)}$  where  $a^*(s) := \arg \max_a Q^\pi(s, a)$  is the optimal action under the agent’s current model  $M_t$ . Deterministic  $\pi^*$  is canonical for finite-MDP RL with isolated optima<sup>[34]</sup>; the perturbative extension to  $\epsilon$ -stochastic and softmax-regularized optima (Section 5.1, full derivation Appendix B.2) covers smooth deviations.

**Value object** Given an objective  $O$ , model  $M_t$ , policy  $\pi$ , and horizon  $N_h$ :  $Q_O(M_t, a; \pi_{\text{cont}}, N_h) = \mathbb{E}[V_O(\tau) \mid M_t, \text{do}(a_t = a), a_{t+1:} \sim \pi_{\text{cont}}]$ . The  $\text{do}(\cdot)$ <sup>[26]</sup> flags intervention on  $a_t$ , not conditioning. We adopt the **one-step improvement** convention  $\pi_{\text{cont}} = \pi_{\text{current}}$  as default: most conservative, comparable across rounds, fixed-point-free. Receding-horizon and Bellman alternatives are in Appendix C.1.

**Bounded value range**  $V_{\max}(M_t) := \max_a Q_O(M_t, a) - \min_a Q_O(M_t, a)$ , finite under bounded reward and finite horizon.

181 **Action gap (isolated optimum)**  $\Delta(a) := Q_O(a^*) - Q_O(a) \in [0, V_{\max}]$ ,  $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) >$   
182 0 whenever the optimum is isolated.

183 **Strategic regret** For policy distribution  $Q$ :  $R(Q) := Q_O(a^*) - \mathbb{E}_{a \sim Q}[Q_O(a)] = \sum_{a \neq a^*} Q(a) \cdot$   
184  $\Delta(a)$ . Three classical forms —  $\mathbb{E}_{\pi^*}[V] - \mathbb{E}_Q[V]$ ,  $V(a^*) - \mathbb{E}_Q[V]$ ,  $\mathbb{E}_{\pi^*}[V - V_Q]$  — coincide under  
185 deterministic  $\pi^*$ .

186 **Total variation and reverse-KL** For finite action measures  $P, Q$  on  $\mathcal{A}$ ,  $\text{TV}(P, Q) :=$   
187  $\frac{1}{2} \sum_a |P(a) - Q(a)|$ ,  $D_{\text{KL}}(P \| Q) := \sum_a P(a) \log \frac{P(a)}{Q(a)}$  with the convention  $0 \log 0 = 0$ .  
188 Our bounds use the *reverse* KL direction (with  $P = \pi^*$ ); the forward direction  $D_{\text{KL}}(Q \| \pi^*)$  is  $+\infty$   
189 whenever  $Q$  has off-optimum mass and is therefore vacuous as a regret coordinate (full direction-  
190 forcing argument in Appendix C.3).

191 **Singular trajectory** The agent operates on a single, non-forkable causal trajectory:  $a_t$  causally  
192 precedes  $o_{t+1}$ , and the model state  $M_t$ 's sufficiency is defined relative to *this* trajectory rather than to  
193 a model-state equivalence class. This is the operational ground for the do-operator above and for the  
194 closed-loop interventional-access argument (Section 5.3).

## 195 4 Main Result

196 We present the four components first as motivation, then state the formal composition theorem, then  
197 unpack what its conclusions mean.

### 198 4.1 The four components

199 Each component supplies a *distinct epistemic property* that the others cannot deliver. The composition  
200 is necessary for the joint properties (rate + persistence + learnability + corrective-action routing), not  
201 for the rate alone. The component-by-component failure-mode analysis is in Appendix A.6 below.

202 **Component 1: Two-gap diagnostic — corrective-action routing** Let  $\Pi$  be the policy class and  
203  $V_O^{\min}$  the trajectory-value threshold above which the objective is met. Define the *satisfaction gap*  
204  $\delta_{\text{sat}} := V_O^{\min} - \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h)$  (goal-feasibility: positive means unmet under the best available  
205 policy) and the *control regret*  $\delta_{\text{regret}} := \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h) - V_O(M_t, \pi_{\text{current}}; N_h) \geq 0$  (policy-  
206 quality). The  $2 \times 2$  cross routes four regimes — *Success* / *Strategy problem* (revise the policy) /  
207 *Capability limit* (revise model, policy class, or horizon) / *Both* — along an axis *orthogonal* to the  
208 exploration-vs-adaptation decompositions of<sup>[5–7]</sup>.

209 **Component 2: Point-mass reverse-KL/TV identity — metric layer on policy space** Un-  
210 der deterministic  $\pi^* = \delta_{a^*}$ , the reverse-KL and total variation collapse to an *exact identity*  
211  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$ , yielding the two-sided regret characterization  $\Delta_{\min}(1 - e^{-K}) \leq$   
212  $R(Q) \leq V_{\max}(1 - e^{-K})$  — *Lipschitz-equivalent* with constants  $\Delta_{\min}/V_{\max}$  and 1, and *coordinate-*  
213 *optimal among bounds depending only on TV*. The identity sits *strictly below* both the Pinsker  $\sqrt{D_{\text{KL}}}/2$   
214 and Bretagnolle–Huber  $\sqrt{1 - e^{-D_{\text{KL}}}}$  envelopes at this corner (Appendix E); the deterministic- $\pi^*$   
215 scope extends perturbatively to  $\epsilon$ -greedy and softmax-regularized optima with quantified corrections  
216 (Section 5.1 + Appendix B.2).

217 **Component 3: Multi-factor strategic tempo with forgetting prerequisite — non-stationarity**  
218 **persistence** A multi-factor strategic tempo composed per revisable policy element of an observation  
219 rate  $\nu$ , an identifiability coefficient  $\iota \in [0, 1]$ , and an update gain  $1 - \lambda$  yields a structural survival  
220 inequality  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_{\Sigma}/R_{\Sigma}$ , the *bottleneck* (per-element  
221 minimum) being the right aggregator since adversarial disturbance concentrates on the weakest  
222 element. A structural-class theorem on the *gain-decay* class  $\mathcal{A}_{\text{decay}}$  (count-accumulating Bayesian  
223 without forgetting, observation-aggregating without restart, gradient-based with vanishing step) shows  
224 every member eventually violates the threshold for any positive disturbance rate (Appendix C.6);  
225 finite-gain mechanisms (constant-step SA, sliding-window, bounded-memory, block-restart) face  
226 *bidirectional* ceilings instead. This *lifts* the tempo result of Lee et al. 2023<sup>[8]</sup> from hyperparameter  
227 optimization to a structural-threshold claim. Full Model ( $\Sigma$ ) and derivation in Section 5.2.

228 **Component 4: Closed-loop interventional access — on-policy learnability** Under (C1) *positivity*  
 229  $(\pi_t(a \mid H_t) \geq p_{\min} > 0)$ , (C2) *sequential ignorability* ( $H_t$  d-separates  $a_t$  from  $o_{t+1}$  in the mutilated  
 230 graph  $G_{\bar{a}_t}$ ), and (C3) *known action mechanism*, the loop’s data-generating process is *interventional*  
 231 — the conditional  $P(o_{t+1} \mid a_t, H_t)$  coincides with the do-distribution  $P(o_{t+1} \mid \text{do}(a_t), H_t)$  on  
 232 the realized-action support (Section 5.3). This makes Component 2’s regret bound *learnable on-*  
 233 *policy* in regimes with sufficient identifiability, with bias from misidentification quantified by  $1 - p_{\text{id}}$   
 234 (Section 5.4). Three regimes<sup>[27]</sup> partition usable strength: A (intervention-rich,  $\iota \approx 1$ ), B (partial), C  
 235 (observation-only).

## 236 4.2 The composition theorem

237 **Theorem 4.1** (Composition: joint guarantees with conditional cumulative reduction). *Let*  
 238  $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$  *be a non-stationary MDP with bounded reward, finite action space, and for each*  
 239 *round  $t$  a deterministic optimum policy  $\pi_t^* = \delta_{a_t^*}$  with action gap  $\Delta_{\min} > 0$ . Suppose the agent*  
 240 *operates on a singular causal trajectory in the sense of Section 3 and updates its policy with per-*  
 241 *element exponential forgetting at rates  $\{\lambda_{ij}\}$ . Let  $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$  be the per-state*  
 242 *reverse-KL coordinate (in the bandit case  $N_h = 1$  the state argument is suppressed). If*

243 **(A1) Metric extension.** *The identity  $\text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$  of Lemma 5.1 is read in*  
 244 *the extended real sense: when  $Q_t(a_t^*(s) \mid s) = 0$ ,  $K_t(s) = +\infty$  and  $1 - e^{-K_t(s)} = 1$  recovers the*  
 245 *trivial bound  $R(Q_t \mid s) \leq V_{\max}$ . No pointwise positivity is required; vanilla deterministic UCB /*  
 246 *UCBVI / UCRL2 deployed at the optimism-argmax are in scope without modification.*

247 **(A2) Multi-factor forgetting prerequisite.** *Within Model  $(\Sigma)$  of Section 5.2, the bottleneck strategic*  
 248 *tempo  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_{\Sigma} / R_{\Sigma}$ .*

249 **(A3) Identifiability regime.** *The action mechanism is known and (C1)–(C3) of Section 5.3 hold; the*  
 250 *agent operates in Regime A or Regime B with per-state argmax-correctness probability  $p_{\text{id}}(s)$ , with*  
 251 *floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ .*

252 **(A4) Two-gap diagnostic.** *The agent applies the satisfaction-gap / control-regret  $2 \times 2$  to route*  
 253 *corrective action.*

254 **(A5) Piecewise-stationary block structure with restart-on-change identity-tight base learner.** *The*  
 255 *time axis partitions into  $B_T + 1$  stationary intervals on which the MDP is stationary and  $\pi_t^*(\cdot \mid s) =$*   
 256  *$\delta_{a_t^*(s)}$  is fixed per state. The base learner is restarted at each block boundary; within each restarted in-*  
 257 *terval of length  $L$  it satisfies  $\sum_{t=1}^L \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{L}$ , where  $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid$*   
 258  *$s_h), Q_t(\cdot \mid s_h))]$  is the occupancy-weighted per-round coordinate. (A non-restarting carryover*  
 259 *variant is plausible but requires bounding cross-block occupancy shift; see Remark in Section 5.)*

260 *Then:*

261 **(i) Per-round regret is two-sided identity-bounded.** *At every visited state,  $\Delta_{\min}(1 - e^{-K_t(s)}) \leq$*   
 262  *$R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$ .*

263 **(ii) Aggregate strategic mismatch is ultimately bounded.** *Under (A2), the modeled mismatch  $\|\delta_{\Sigma}\|$  is*  
 264 *ultimately bounded within  $R_{\Sigma}^* = \rho_{\Sigma} / \mathcal{T}_{\Sigma}^{\text{bn,ss}}$ .*

265 **(iii) The KL coordinate is computable with controlled bias.**  *$-\log Q_t(a_{\text{ag},t}^*(s) \mid s)$  is computable*  
 266 *directly from  $Q_t$ . Per-state bias:  $\mathbb{E}[\|\hat{D}_t(s) - D_t^{\text{true}}(s)\|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$  under  $Q_t(\cdot \mid s) \geq q_0$*   
 267 *at the two argmax candidates. Vanishes in Regime A.*

268 **(iv) The  $2 \times 2$  cell identifies the corrective-action class.**  *$(\delta_{\text{sat}}, \delta_{\text{regret}})$  routes to revise-policy / revise-*  
 269 *model-class-or-horizon / revise-objective.*

270 **(v) Trajectory-level blockwise dynamic regret.** *Under (A5),  $\mathbb{E}[\text{DynReg}(T)] :=$*   
 271  *$\mathbb{E}\left[\sum_{t=1}^T (V_t^{\pi_t^*} - V_t^{Q_t})\right] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) \cdot T$ .*

272 The proof composes the per-round identity (i) with the simulation lemma<sup>[38–41]</sup>, a block decomposition  
 273 at optimum-change events, Cauchy–Schwarz across the  $B_T + 1$  blocks, and the bias bound from (iii).  
 274 Full proof in Appendix A; the four key lemmas are surfaced in Section 5.

### 4.3 Unpacking the conclusions

Theorem 4.1 asserts a non-stationarity-aware convergence theory with three jointly-novel properties: *non-stationarity persistence* (via (ii)/(A2)), *explicit metric structure on policy space* (via (i)/Component 2), *on-policy learnability* (via (iii)/(A3)). Conclusion (iv) is the connective tissue; conclusion (v) the cumulative rate from composing per-round metric with base-learner stationary rate.

*The per-round coordinate is exact, not an inequality.* Under deterministic  $\pi^*$ , the identity-bound (i) is  $\Delta_{\min}/V_{\max}$ -Lipschitz-equivalent to regret with both endpoints achieved on extremal value landscapes — *coordinate-optimal among bounds depending only on TV*. Strictly sharper than Pinsker  $\sqrt{D_{\text{KL}}/2}$  (exact, not approximate) and Bretagnolle–Huber  $\sqrt{1 - e^{-D_{\text{KL}}}}$  at this corner ( $x < \sqrt{x}$  on  $(0, 1)$ ).

*Cumulative rate: piecewise-stationary  $B_T$  family by direct aggregation, continuous-variation  $V_T$  family by MASTER wrapping.* The lifted UCRL2/UCBVI rate  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  (Appendix D) sits in the restart-on-change family of<sup>[1]</sup>; via<sup>[2]</sup>’s MASTER black-box, the same identity-routed base learner achieves Best-of-Both-Worlds  $\tilde{O}(\min\{\sqrt{(B_T + 1)T}, (V_T + 1)^{1/3}T^{2/3}\})$  automatically — matching<sup>[3]</sup>’s near-optimal  $V_T$  exponent (parameter  $(B_T$  vs  $V_T)$ ,  $T$ -scaling,  $S$ ,  $A$ -scaling differ from Mao’s RestartQ-UCB; full development in Section 6 + (A5’)-BoBW Remark in Appendix A).

*The bias term in (v) is a value-coordinate misidentification penalty.* The second term  $V_{\max}N_h(1 - p_{\text{id}}) \cdot T$  is the expected value gap between true and agent-identified optima,  $V_{\max}$ -bounded on the misidentification event of probability  $1 - p_{\text{id}}$  aggregated over  $N_h$  horizon steps. Vanishes in Regime A; in Regime B the dominant rate term is the metric one when  $1 - p_{\text{id}}$  is small relative to  $\sqrt{(B_T + 1)T}$ ; in Regime C saturates at the trivial  $V_{\max}N_h \cdot T$  envelope (Appendix D flags Regime C out-of-scope). *Strict assumption-set weakening:* (v) does not require Lemma 4’s two-point support  $Q_t \geq q_0$  — that condition is needed only for conclusion (iii)’s diagnostic computability.

*The bundle is compatible, not integrated.* (v)’s rate goes through (A5) and (i) alone — Component 2 supplies the metric, (A5) the base-learner regret, the  $\sqrt{(B_T + 1)T}$  aggregation falls out. The other components carry the *other* guarantees: Component 1 the routing (iv); Component 3 the persistence (ii) (without it,  $K_t$  may itself drift unboundedly under  $\mathcal{A}_{\text{decay}}$ ); Component 4 the on-policy learnability via (iii) (without it, (v) becomes vacuous in Regime C). The horizon factor  $N_h$  in (v) is the linear-in- $N_h$  simulation-lemma penalty, sharper than the  $N_h^2$  of TRPO-style worst-state bounds<sup>[42]</sup>. Component-by-component failure-mode analysis in Appendix A.6.

## 5 Mechanism

We surface four key lemmas and chain them through a block decomposition. Algebra is deferred to the appendices.

### 5.1 Key Lemma 1: Point-mass reverse-KL/TV identity

**Lemma 5.1** (Point-mass reverse-KL/TV identity). *For deterministic  $\pi^* = \delta_{a^*}$  (per visited state, with  $a^* = a^*(s)$  and  $Q = Q(\cdot \mid s)$ ) and any policy  $Q$ ,  $D_{\text{KL}}(\pi^* \parallel Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q))$ , equivalently  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \parallel Q)}$ . The identity is read in the extended real sense: when  $Q(a^*) = 0$  both sides equal 1, with the natural convention  $D_{\text{KL}} = +\infty$  and  $e^{-\infty} = 0$  — the identity holds for all policies  $Q$ , including Diracs at suboptimal actions.*

*Intuition.* Both  $D_{\text{KL}}$  and TV collapse cleanly under the point-mass — every  $a \neq a^*$  term vanishes by  $0 \log 0 = 0$ . Substituting into BH gives  $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ , strict on  $D_{\text{KL}} > 0$ : the identity is the *exact value* at this corner rather than BH-at-equality. Extends perturbatively to  $\epsilon$ -greedy and softmax-regularized optima with  $O(\epsilon \log(1/\epsilon))$  and  $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$  corrections under full-support  $Q \geq q_0$  (Appendix B.2). Full proof in Appendix B.

### 5.2 Key Lemma 2: Forgetting prerequisite

The strategic mismatch state  $\delta_{\Sigma} = (\delta_{ij})_{(i,j) \in E}$  in the Euclidean norm  $\|\delta_{\Sigma}\|^2 = \sum_{(i,j)} \delta_{ij}^2$  evolves in discrete time as  $\mathbb{E}[\Delta \delta_{ij} \mid \delta_{ij}] = -\alpha_{ij} \delta_{ij} + w_{ij}$ ,  $|w_{ij}| \leq \rho_{\Sigma}/|E|^{1/2}$ , where  $\alpha_{ij} \geq 0$  is a per-element correction rate,  $w_{ij}$  a bounded disturbance with budget  $\sum w_{ij}^2 \leq \rho_{\Sigma}^2$ , and  $R_{\Sigma}$  the operative

norm-radius beyond which tracking is lost. Beta-Bernoulli edge updates and Robbins–Monro gradient updates both instantiate this *Model* ( $\Sigma$ ).

**Lemma 5.2** (Multi-factor forgetting prerequisite). *Within Model ( $\Sigma$ ) with diagonal per-element correction rates  $\alpha_{ij} = \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij}$  (per-element observation rate  $\nu_{ij} \in [0, 1]$ , regime-adjusted identifiability  $\iota_{ij} \in [0, 1]$ , edge-update gain  $\eta_{\text{edge},ij} > 0$ ), under per-element exponential forgetting at rate  $\lambda_{ij} \in (0, 1)$  giving steady-state  $\eta_{\text{edge},ij}^{\text{ss}} \approx 1 - \lambda_{ij}$ , ultimate boundedness of  $\|\delta_\Sigma\|$  within  $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$  is sufficient under  $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} (\nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})) > \rho_\Sigma / R_\Sigma$ . The threshold is sharp inside the diagonal sector model: when the inequality reverses, an adversarial disturbance concentrating on the bottleneck element drives  $\|\delta_\Sigma\|$  beyond  $R_\Sigma$  within the modeled dynamics.*

*Intuition (why bottleneck, not sum).* Bounding  $\sum_{(i,j)} \alpha_{ij} \delta_{ij}^2 \geq c \|\delta_\Sigma\|^2$  uniformly forces  $c = \min \alpha_{ij}$  — adversarial disturbance concentrating on the weakest element saturates at the bottleneck (the same effect makes per-dimension persistence conditions sharper than scalar conditions in adaptive control<sup>[43]</sup>). Ultimate boundedness then follows from Khalil’s sector-Lyapunov template; full proof in Appendix B, structural-class theorem on  $\mathcal{A}_{\text{decay}}$  in Appendix C.6.

*ProST as a sector-level reduction.* The block-restart structure of ProST<sup>[8]</sup> recovers as the *impulsive limit* of Model ( $\Sigma$ ): between updates the mismatch evolves under disturbance alone, and each update contracts the mismatch by per-update gain  $\gamma$ . The Hespanha–Liberzon–Teel reverse-ADT framework<sup>[44]</sup> gives the threshold (full development in Appendix B.4); under uniform schedules this recovers ProST’s  $K/T$  form with the impulse gain  $\gamma$  made explicit. The reframe lifts ProST’s *tempo-as-hyperparameter* framing into *tempo-as-stability-margin* — the longest inter-update gap, not the average, sets the threshold, exposing the tradeoff between update frequency and per-update impulse strength at the sector-Lyapunov level.

### 5.3 Key Lemma 3: Closed-loop interventional access

**Lemma 5.3** (Loop generates interventional samples). *Let  $H_t$  denote the agent’s information state at round  $t$  (model state  $M_t$ , history of observations and actions). Under (C1) positivity —  $\pi_t(a \mid H_t) \geq p_{\min} > 0$  on the support of  $H_t$  for the actions of interest; (C2) sequential ignorability —  $H_t$   $d$ -separates  $a_t$  from  $o_{t+1}$  in the mutilated graph  $G_{\overline{a_t}}$  (equivalently,  $a_t \perp\!\!\!\perp Y^{(a_t)} \mid H_t$  in potential-outcome notation); (C3) known action mechanism — the behavior policy  $\pi_t(a \mid H_t)$  is known: executed actions are interventions relative to the environment’s transition kernel, and the loop generates samples from  $P(o_{t+1} \mid \text{do}(a_t = a), H_t)$  — i.e., Pearl Level-2 interventional kernels conditional on the decision context. The substantive content for the identification claim is (C2); under the singular-trajectory + agent-as-policy architecture of Section 3, (C1) reduces to realized-action positivity (automatic) and (C3) to agent-policy-queryability (architectural). (C1) and (C3) are stated explicitly for forward-compatibility with off-policy / counterfactual extensions.*

*Intuition.* By temporal ordering,  $a_t$  causally precedes  $o_{t+1}$ . Under (C1)–(C3) the conditional  $P(o_{t+1} \mid a_t, H_t)$  coincides with the interventional  $P(o_{t+1} \mid \text{do}(a_t), H_t)$  on the support where  $\pi_t(a_t \mid H_t) > 0$  — the loop generates conditional-Level-2 samples regardless of how the agent updates internally. Three regimes<sup>[27]</sup> partition usable strength: A ( $\iota \approx 1$ , do-effects identified cleanly), B ( $\iota \in (0, 1)$ , bias  $\propto 1 - \iota$ ), C (observation-only, not on-policy realizable; observer-on-sub-agent extensions in Appendix C.8). Components 2 and 4 are jointly load-bearing — neither alone makes the bound learnable.

### 5.4 Key Lemma 4: Bias bound under partial identifiability

**Lemma 5.4** (Bias bound). *Assume  $Q_t(a \mid s) \geq q_0 > 0$  for both  $a = a_{\text{ag},t}^*(s)$  (the agent’s identified optimum) and  $a = \tilde{a}_t^*(s)$  (the true optimum). Let  $\hat{D}_t(s) := -\log Q_t(a_{\text{ag},t}^*(s) \mid s)$  and  $D_t^{\text{true}}(s) := -\log Q_t(\tilde{a}_t^*(s) \mid s)$ . Then  $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$ , and taking expectations:  $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$ , where  $p_{\text{id}}(s) := \Pr[a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)]$ .*

*Intuition.* Indicator decomposition: zero on the matching event; on the misidentification event both  $\log Q_t$  values lie in  $[\log q_0, 0]$  so their absolute difference is at most  $\log(1/q_0)$ . Bias scales linearly in misidentification mass — vanishes in Regime A, controlled in Regime B, saturates at  $\log(1/q_0)$  in



372 Regime C. The two-point support condition (at the two argmax candidates) is strictly weaker than the  
 373 perturbative identity’s full-support condition (Section 5.1). Full proof in Appendix B.

## 374 5.5 Combining the lemmas: cumulative regret in four steps

375 *Step 1 — Per-round bound via the simulation lemma.* The performance-difference lemma<sup>[38–41]</sup> gives,  
 376 for any policies  $\pi_t^*$  and  $Q_t$ ,  $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$ , where  $\overline{\text{TV}}_t$  is the occupancy-  
 377 weighted TV of (A5). By Key Lemma 1 applied per state,  $\text{TV}(\pi_t^*, Q_t) = 1 - e^{-K_t(s)}$  — sharper  
 378 than Pinsker, preserved under occupancy aggregation. The  $N_h$  is the linear-in- $N_h$  simulation-lemma  
 379 penalty, sharper than the  $N_h^2$  of TRPO-style worst-state bounds<sup>[42]</sup>.

380 *Step 2 — Block decomposition.* Partition  $[1, T]$  at  $(P, r)$ -change events into blocks of lengths  $\Delta_i$  (the  
 381 kernel-stationary-segment boundaries of Section 3). (A5) — restart-on-change base learner — gives  
 382 within-block  $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$ .

383 *Step 3 — Cauchy–Schwarz across  $B_T + 1$  blocks.*  $\sum_i \sqrt{\Delta_i} \leq \sqrt{(B_T + 1)T}$ , giving  
 384  $\sum_t \mathbb{E}[V_{\max} N_h \overline{\text{TV}}_t] \leq 2cV_{\max} N_h \sqrt{(B_T + 1)T}$ .

385 *Step 4 — Misidentification penalty.* Steps 1–3 give the rate term against the agent’s *identified* optimum  
 386  $\pi_{\text{ag},t}^*$  (what (A5) delivers); (v) compares to the *true* optimum  $\tilde{\pi}_t^*$ . Applying the simulation lemma to  
 387 the two point-mass policies, each per-step bracket reduces to  $Q_h^{\tilde{\pi}^*}(s_h, \tilde{a}^*) - Q_h^{\pi_{\text{ag},t}^*}(s_h, a_{\text{ag}}^*) \in [0, V_{\max}]$   
 388 on the misidentification event and zero on the matching event. Aggregated over  $N_h$  horizon steps  
 389 with floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ , the per-round penalty is at most  $V_{\max} N_h (1 - p_{\text{id}})$  — value-coordinate  
 390 throughout, vanishing in Regime A. The support condition  $Q_t \geq q_0$  is needed only for conclusion  
 391 (iii)’s diagnostic readout (Lemma 4), not for the rate term.

392 Combining:  $\mathbb{E}[\text{DynReg}(T)] \leq 2cV_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) \cdot T$ , with Cesàro tracking  
 393  $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) \rightarrow 0$  when  $B_T = o(T)$  and  $p_{\text{id}} \rightarrow 1$ .

394 In the bandit special case the rate sharpens logarithmically; the non-restarting carryover variant of  
 395 (A5) is deferred. Full proof of (v) in Appendix A; bandit-case sharpening + algorithmic instantiation  
 396 in Appendix D.

## 397 6 Conclusion

398 The four-component composition delivers three properties no published strand has individually:  
 399 *handles non-stationarity* (strategic-tempo persistence), *explicit metric structure on policy space*  
 400 (point-mass identity), *on-policy learnable* (closed-loop interventional access). The cumulative re-  
 401 regret rate is the Best-of-Both-Worlds form  $\tilde{O}(\min\{V_{\max} N_h \sqrt{(B_T + 1)T}, V_{\max} N_h (V_T + 1)^{1/3} T^{2/3}\})$   
 402 — automatically adapting between piecewise-stationary  $B_T$  and continuous variation  $V_T$  via Wei-  
 403 Luo MASTER wrapping (Appendix A (A5’)-BoBW Remark), with per-round identity supplying a  
 404 coordinate that’s exact at the deterministic- $\pi^*$  corner.

405 **On structural unifiability of the variation regimes** The  $B_T$  and  $V_T$  regimes share the per-round co-  
 406 ordinate; they differ in aggregation strategy (block-Cauchy–Schwarz vs. adaptive-window MASTER).  
 407 The  $V_T$  exponent matches<sup>[31]</sup>’s near-optimal rate, and<sup>[28]</sup>’s lower bound applies at the deterministic- $\pi^*$   
 408 corner — so the per-round identity sharpens constants and per-round form, not the  $V_T$  exponent. Two  
 409 refinements remain open: an *agent-experienced* variation budget  $V_T^{\text{eff}, \Sigma}$  counting only the times  $\|\delta_{\Sigma}\|$   
 410 exits its ultimate-bounded region, and an *identity-coordinate* variation  $V_T^{(K)} := \sum_t |K_t - K_{t-1}|$  —  
 411 both bounded above by environment-side  $V_T$  but typically much smaller in absorbing regimes where  
 412 the strategic substate damps drift. The result would explain why agents in slow-drift environments  
 413 outperform  $B_T$ -pessimistic predictions.

414 **On coupled-goal architectures** (C2)’s sequential-ignorability presupposes architectural separation  
 415 between model state  $M_t$  and goal state. Goal-conditioned LLM policies violate this: the goal  
 416 influences both action selection and environment response via shared latent context (user intent,  
 417 dialogue history, task framing), so  $H_t$  does not block the goal-mediated path from  $a_t$  to  $o_{t+1}$  — (C2)  
 418 fails.<sup>[45]</sup>’s conditional-independence relaxation is a starting point.

419 **Open scope questions** *Optimal  $N_h$ ,  $S$ ,  $A$  dependence:* the lifted UCRL2/UCBVI rate  
 420  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  could shave one  $\sqrt{N_h}$  factor via Bernstein-type concentration<sup>[41]</sup>. *Joint*  
 421 *uniqueness:* metric layer is uniquely reverse-KL up to scaling under chain-rule additivity (Theorem C.3); we do not claim joint uniqueness across all three properties. *Strategic-tempo beyond Model*  
 422 *( $\Sigma$ ):* the bottleneck threshold is sharp inside the diagonal sector model; non-diagonal correction is  
 423 open in the adaptive-control-RL bridge.  
 424

425 **Practitioner takeaway** Use the aggregate necessary-condition  $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$  as a fail-fast  
 426 pre-check; estimate  $\rho_{\Sigma}/R_{\Sigma}$  from observed variation<sup>[1,2,37]</sup>; route corrective action via the  $2 \times 2$  diagnos-  
 427 tic; apply the per-round identity wherever deterministic- $\pi^*$  scope holds. Regime C (observation-only)  
 428 flags itself as out-of-scope.

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## A Proof of the Composition Theorem

This appendix gives the full proof of Theorem 4.1. The proof composes the four key lemmas of Section 5 (full proofs of those in Appendix B) with a block decomposition at optimum-change events plus Cauchy–Schwarz across blocks.

**Notation** Throughout this section:  $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$  is the per-state reverse-KL coordinate;  $d_h^{Q_t}(\cdot \mid s_0)$  is the round- $t$  learner-induced state distribution at horizon step  $h$  starting from  $s_0$ ;  $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid s_h), Q_t(\cdot \mid s_h))]$  is the occupancy-weighted per-round coordinate of (A5);  $B_T = |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$  counts piecewise-stationary segment boundaries (per Section 3); the partition is  $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$  with block  $i$  length  $\Delta_i := \tau_{i+1} - \tau_i$ .

### 550 A.1 Proof of (i) — Per-round identity-bounded regret

551 For deterministic  $\pi_t^* = \delta_{a_t^*(s)}$  and any  $Q_t(\cdot | s)$ , Key Lemma 1 (Lemma 5.1, full proof in Ap-  
 552 pendix B.1) gives the exact identity  $\text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot | s)) = 1 - e^{-K_t(s)}$  in the extended real sense  
 553 (when  $Q_t(a_t^*(s) | s) = 0$ ,  $K_t(s) = +\infty$  and both sides equal 1, recovering the trivial bound).  
 554 Composed with the per-state TV-regret bounds (full proofs in Appendix B.1 and Appendix C.4):  
 555  $\Delta_{\min} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot | s)) \leq R(Q_t | s) \leq V_{\max} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot | s))$ , substituting the identity  
 556 yields conclusion (i):  $\Delta_{\min}(1 - e^{-K_t(s)}) \leq R(Q_t | s) \leq V_{\max}(1 - e^{-K_t(s)})$ .  $\square$

### 557 A.2 Proof of (ii) — Aggregate mismatch ultimate boundedness

558 Under (A2), Key Lemma 2 (Lemma 5.2, full proof in Appendix B.3) applied to the diagonal sector  
 559 model with  $\alpha_{ij}^{\text{ss}} = \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$  gives ultimate boundedness of  $\|\delta_{\Sigma}\|$  within  $R_{\Sigma}^* = \rho_{\Sigma} / \mathcal{T}_{\Sigma}^{\text{bn,ss}}$ .  $\square$

### 560 A.3 Proof of (iii) — KL coordinate bias bound

561 Key Lemma 4 (Lemma 5.4, full proof in Appendix B.6) gives the per-state bias bound directly, under  
 562 (A3)’s identifiability assumption (which gives  $p_{\text{id}}(s)$  for each  $s$ ) and the two-point support condition  
 563  $Q_t(\cdot | s) \geq q_0$  at  $a_{\text{ag},t}^*(s)$  and  $\tilde{a}_t^*(s)$ .  $\square$

### 564 A.4 Proof of (iv) — $2 \times 2$ corrective-action routing

565 (A4) states the agent applies the satisfaction-gap / control-regret  $2 \times 2$  (Section 4.1 Component 1).  
 566 Conclusion (iv) is a direct restatement of the  $2 \times 2$  cell partition. The decomposition is convention-  
 567 dependent (the values of  $\delta_{\text{sat}}$  and  $\delta_{\text{regret}}$  depend on the continuation convention; see Appendix C.1 for  
 568 monotonicity across C1/C2/C3); the  $2 \times 2$  structure is preserved across all three conventions.  $\square$

### 569 A.5 Proof of (v) — Trajectory-level cumulative regret

570 We chain Key Lemmas 1 and 4 with the simulation lemma and a Cauchy–Schwarz block argument.  
 571 The proof is in four steps; the high-level sketch is in Section 5.5.

572 *Step 1: Simulation / performance-difference lemma.* For any two policies  $\pi$   
 573 and  $\pi'$ , any state  $s$ , and finite-horizon non-discounted MDP with horizon  $N_h$ ,  
 574 the standard performance-difference lemma<sup>[38–41]</sup> gives  $V_t^{\pi'}(s) - V_t^{\pi}(s) =$   
 575  $\mathbb{E}\left[\sum_{h=0}^{N_h-1} \left(\mathbb{E}_{a' \sim \pi'(\cdot | s_h)}[Q_h^{\pi'}(s_h, a')] - Q_h^{\pi}(s_h, a_h)\right) \mid s_0 = s, a_h \sim \pi\right]$ . Each per-step bracket  
 576 is bounded by value range times TV:  $|\mathbb{E}_{a' \sim \pi'}[Q_h^{\pi'}] - \mathbb{E}_{a \sim \pi}[Q_h^{\pi}]| \leq V_{\max} \cdot \text{TV}(\pi'(\cdot |$   
 577  $s_h), \pi(\cdot | s_h))$ . Applied at round  $t$  with  $\pi' = \pi_t^*$ ,  $\pi = Q_t$ :  $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq$   
 578  $V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}(\cdot | s_0)}[\text{TV}(\pi_t^*(\cdot | s_h), Q_t(\cdot | s_h))] = V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$ . The horizon factor  $N_h$   
 579 here is the *linear-in- $N_h$*  simulation-lemma penalty — sharper than the  $N_h^2$  of TRPO-style worst-state  
 580 bounds<sup>[42]</sup>, which bound max-state TV rather than occupancy-weighted TV.

581 *Step 2: Per-state TV identity along the trajectory.* By the deterministic-per-state optimum scope of  
 582 Theorem 4.1 ( $\pi_t^*(\cdot | s) = \delta_{a_t^*(s)}$ ), at every visited state Key Lemma 1 gives  $\text{TV}(\pi_t^*(\cdot | s), Q_t(\cdot |$   
 583  $s)) = 1 - Q_t(a_t^*(s) | s) = 1 - e^{-K_t(s)}$ . Summing over the trajectory and taking expectations:  
 584  $\overline{\text{TV}}_t = \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}}[1 - e^{-K_t(s_h)}]$ . The per-state KL/TV identity is preserved underneath  
 585 the occupancy aggregation; the per-round coordinate is sharper than Pinsker/BH at every visited state.

586 *Step 3: Block-sum + Cauchy–Schwarz across blocks.* Within each block  $[\tau_i, \tau_{i+1})$  of length  
 587  $\Delta_i$ , (A5)’s restart-on-change identity-tight base learner gives  $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$ . Sum-  
 588 ming over the  $B_T + 1$  blocks:  $\sum_{t=1}^T \mathbb{E}[\overline{\text{TV}}_t] = \sum_{i=0}^{B_T} \sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c \sum_{i=0}^{B_T} \sqrt{\Delta_i}$ .  
 589 Cauchy–Schwarz gives  $\sum_{i=0}^{B_T} \sqrt{\Delta_i} \leq \sqrt{(B_T + 1) \cdot \sum_i \Delta_i} = \sqrt{(B_T + 1)T}$ . Multiplying by  
 590  $V_{\max} N_h$ :  $\sum_{t=1}^T \mathbb{E}[V_{\max} N_h \overline{\text{TV}}_t] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T}$ . This is the rate term.

591 *Step 4: Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap.* Steps  
 592 1–3 implicitly compared the agent’s policy  $Q_t$  to the agent’s identified-optimum policy  $\pi_{\text{ag},t}^*(\cdot |$

593  $s) := \delta_{a_{\text{ag},t}^*(s)}$  (since (A5)’s base-learner guarantee is what the agent achieves on its identified  
594 objective). The trajectory-level dynamic regret in (v) compares to the *true* optimum  $\tilde{\pi}_t^*(\cdot | s) := \delta_{\tilde{a}_t^*(s)}$ .  
595 Decompose:  $V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{Q_t}(s_0) = [V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{\pi_{\text{ag},t}^*}(s_0)] + [V_t^{\pi_{\text{ag},t}^*}(s_0) - V_t^{Q_t}(s_0)]$ . Steps  
596 1–3 bound the second bracket. The first bracket is the misidentification penalty. Apply the simulation  
597 lemma to the two point-mass policies  $(\tilde{\pi}_t^*, \pi_{\text{ag},t}^*)$ : each per-step bracket reduces to  $Q_h^{\tilde{\pi}_t^*}(s_h, \tilde{a}_t^*(s_h)) -$   
598  $Q_h^{\pi_{\text{ag},t}^*}(s_h, a_{\text{ag},t}^*(s_h)) \in [0, V_{\max}] \cdot \mathbf{1}[\tilde{a}_t^*(s_h) \neq a_{\text{ag},t}^*(s_h)]$ . Lower bound by optimality of  $\tilde{a}^*$ ; upper  
599 bound by the cumulative-Q range with the indicator on the misidentification event. Taking expectations,  
600 with the per-state floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ :  $\mathbb{E}[V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{\pi_{\text{ag},t}^*}(s_0)] \leq V_{\max} \cdot N_h \cdot (1 - p_{\text{id}})$ . Summing  
601 over  $T$  rounds:  $V_{\max} \cdot N_h \cdot (1 - p_{\text{id}}) \cdot T$ . This is the bias term, in *value coordinates* throughout — no  
602 detour through the KL coordinate. Key Lemma 4’s KL-readout bound is the right tool for conclusion  
603 (iii)’s diagnostic computability; for the value-coordinate misidentification penalty here, the indicator  
604 decomposition is direct and tighter (the support condition  $Q_t \geq q_0$  at the two argmax candidates is  
605 *not* required for this term — only conclusion (iii) needs it).

606 *Combining.* Adding Steps 1–4:  $\mathbb{E}[\text{DynReg}(T)] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) T$ ,  
607 which is conclusion (v).  $\square$

608 **Cesàro tracking corollary** Dividing through by  $T$ :  $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) \leq$   
609  $2c V_{\max} N_h \sqrt{(B_T + 1)/T} + V_{\max} N_h (1 - p_{\text{id}})$ . As  $T \rightarrow \infty$  with  $B_T = o(T)$  and  $p_{\text{id}} \rightarrow 1$   
610 (Regime A), the right side tends to zero — the agent’s average value over the trajectory tracks the  
611 moving optimum’s average value. In Regime B the second term is constant; tracking is only up to a  
612  $V_{\max}(1 - p_{\text{id}})$  residual.  $\square$

613 **Bandit case sharpening (Remark)** In the bandit special case ( $N_h = 1$ , so  $\overline{\text{TV}}_t = \mathbb{E}[1 - e^{-K_t}]$ ),  
614 Thompson sampling and UCB (Appendix D) satisfy (A5) with the *stronger logarithmic* per-block rate.  
615 By Lattimore and Szepesvári 2020<sup>[34]</sup> Theorem 7.1 (or [? auer-cesa-bianchi-fischer-2002-finitetime]),  
616  $\mathbb{E}[N_a(L)] = O(\log L / \Delta_a^2)$  per suboptimal arm  $a$ ; aggregating over arms gives  $\mathbb{E}[1 - e^{-K_t}] =$   
617  $\mathbb{E}[1 - Q_t(a^*)] = O(\log t / (t \Delta_{\min}^2))$  per stationary block. Substituting and summing gives cumulative  
618 regret  $O(V_{\max}(B_T + 1) \log(T/(B_T + 1)) / \Delta_{\min}^2)$  — sharper than the worst-case  $\sqrt{(B_T + 1)T}$  when  
619  $\Delta_{\min}$  is bounded away from zero. The framework’s  $V_{\max} \cdot \text{TV}$  chain is structurally one factor of  $\Delta_{\min}$   
620 looser than direct gap-aware UCB analysis (which gives  $O(\log T / \Delta_{\min})$  cumulative regret per block  
621 by weighting each suboptimal pull by  $\Delta(a)$  rather than  $V_{\max}$ ); this is a feature of the unification, not a  
622 bug. For  $N_h > 1$  MDPs, UCRL2 and UCBVI provide the natural occupancy-weighted instantiation  
623 (Appendix D) with  $c$  of order  $\tilde{O}(\sqrt{N_h SA})$ , lifting cleanly to  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  — same  
624 piecewise-stationary  $B_T$  family as restart-on-change variants of<sup>[1]</sup>.

625 **Best-of-Both-Worlds non-stationarity extension (Remark on (A5’))** Conclusion (v) above uses  
626 the restart-on-change form of (A5). The per-round identity route is *stationarity-agnostic* — Steps 1,  
627 2, 4 above hold per round regardless of cross-round dynamics; only Step 3’s block-Cauchy–Schwarz  
628 aggregator is piecewise-stationary-specific. Replacing (A5) by

629 **(A5’) MASTER-wrapped base learner.** The base learner is the Wei and Luo 2021<sup>[2]</sup>  
630 MASTER black-box wrapping of any base algorithm satisfying their Assumption 1  
631 (a near-stationary regret guarantee with  $C(t) = c_1 t^p + c_2$  for  $p \in [1/2, 1)$ , plus an  
632 auxiliary-quantity output condition; UCRL2, UCBVI, and Thompson sampling all  
633 qualify), giving stationary regret  $\sum_{t \in W} \mathbb{E}[\text{TV}_t] \leq 2c \sqrt{|W|}$  within each MASTER-  
634 selected window  $W$ .

635 substitutes Step 3’s block-Cauchy–Schwarz aggregator with MASTER’s adaptive-window aggrega-  
636 tor (parallel base-learner instances at exponentially-spaced window sizes  $W_k = 2^k$ , online meta-  
637 learner over the  $\log T$  instances). The wrapping is mechanical: the per-round identity (Steps 1–2)  
638 is preserved unchanged at each round; the misidentification penalty (Step 4) is round-additive and  
639 rides through; only the aggregator changes. The resulting rate, by Wei-Luo Theorem 2 applied  
640 with our framework’s  $C(t) = 2c V_{\max} N_h \sqrt{t}$  at the boundary  $p = 1/2$ , is  $\mathbb{E}[\text{DynReg}(T)] \leq$   
641  $\tilde{O}\left(V_{\max} N_h \cdot \min \left\{ \sqrt{(B_T + 1)T}, (V_T + 1)^{1/3} T^{2/3} \right\}\right) + V_{\max} N_h (1 - p_{\text{id}}) T$ , automatically —

without prior knowledge of  $B_T$  or  $V_T$ . The  $V_T^{1/3}T^{2/3}$  exponent matches<sup>[3]</sup>’s near-optimal RestartQ-UCB rate and is information-theoretically near-optimal: the<sup>[28]</sup> lower bound  $\Omega(K^{1/3}V_T^{1/3}T^{2/3})$  for non-stationary stochastic bandits applies *also at the deterministic- $\pi$  corner\** (their construction is a Bernoulli MAB with dynamic oracle playing the per-round argmax — deterministic-per-round optimum, identical to our canonical scope), so the per-round identity sharpens *constants and per-round form*, not the  $V_T$  exponent. The two regimes are *not* structurally distinct: they share the per-round coordinate  $1 - e^{-K_t}$  and differ only in the aggregation strategy.  $\square$

## A.6 Component-by-component failure-mode analysis

This composition is a *bundle of compatible guarantees* (each conclusion supported by its own assumptions), not a single integrated theorem in which every component is necessary for the rate. The rate (v) goes through (A5) and (i) alone; the joint properties — rate plus persistence plus learnability plus corrective-action routing — require all four. The failure modes component-by-component:

**Without Component 2 (point-mass identity)** The per-round bound is Pinsker  $V_{\max}\sqrt{D_{\text{KL}}/2}$  or BH  $V_{\max}\sqrt{1 - e^{-D_{\text{KL}}}}$  — both strictly looser at this corner; Pinsker is vacuous for  $D_{\text{KL}} > 2$ . The metric layer loses *exactness*, and the “behavior-cloning loss against optimal trajectory” interpretation of the cumulative coordinate vanishes.

**Without Component 3 (strategic tempo)** The agent has no persistence guarantee. By the structural-class theorem on  $\mathcal{A}_{\text{decay}}$  (Appendix C.6), every count-accumulating update without forgetting eventually violates the threshold for any positive disturbance; the modeled mismatch  $\|\delta_{\Sigma}\|$  grows unboundedly. The per-round bound (i) still holds, but  $K_t(s)$  may itself drift — there is no guarantee the per-round coordinate stays bounded.

**Without Component 4 (loop-Level-2)** The bound’s RHS is computable from  $Q_t$ , but its interpretation as regret-against-true-optimum requires identifying  $a_t^*(s)$  — an interventional query under the environment’s transition kernel. In Regime C ( $\iota \approx 0$ ), the misidentification penalty saturates at  $V_{\max}$  per state per step, contributing  $V_{\max}N_h \cdot T$  to cumulative regret — the rate (v) becomes vacuous.

**Without Component 1 (two-gap diagnostic)** The regret signal  $\delta_{\text{regret}}$  alone does not distinguish “policy revision is the right move” from “the goal is unattainable from  $\bar{M}_t$ .” An agent might persist in policy revision when the model, policy class, or horizon is the actual problem (Capability-limit cell), or revise the goal when policy revision would have sufficed.

So while the rate (v) goes through (A5) and (i) alone, the *bundle* — rate plus persistence plus learnability plus corrective-action routing — requires all four. Each drop is a concrete loss of an epistemic property that the field treats as separate.

## B Proofs of the Key Lemmas

This appendix gives full proofs of the four key lemmas surfaced in Section 5, plus the perturbative extension and the ProST sector-level reduction.

### B.1 Proof of Key Lemma 1 (Point-mass reverse-KL/TV identity)

For deterministic  $\pi^* = \delta_{a^*}$  and any policy  $Q$  (extending  $D_{\text{KL}} = +\infty$  when  $Q(a^*) = 0$ , which gives  $1 - e^{-D_{\text{KL}}} = 1 = \text{TV}$  trivially):

The reverse-KL collapses under the point-mass:  $D_{\text{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)} = \log \frac{1}{Q(a^*)} = -\log Q(a^*)$ , where the convention  $0 \log 0 = 0$  disposes of the  $a \neq a^*$  terms. Combined with the textbook total-variation calculation  $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| = \frac{1}{2} (|1 - Q(a^*)| + \sum_{a \neq a^*} Q(a)) = \frac{1}{2} ((1 - Q(a^*)) + (1 - Q(a^*))) = 1 - Q(a^*)$ , we have  $-\log Q(a^*) = -\log(1 - \text{TV})$ . Solving for TV gives  $\text{TV} = 1 - e^{-D_{\text{KL}}}$ .  $\square$

685 **Strict-improvement substitution** Substituting the identity into the Bretagnolle–Huber inequality  
686  $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$  at  $P = \delta_{a^*}$  yields  $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ , which holds *strictly*  
687 on  $D_{\text{KL}} > 0$  since  $x < \sqrt{x}$  on  $(0, 1)$ . The identity therefore supplies the *exact* TV value at the  
688 deterministic- $\pi^*$  corner and lies strictly below the BH envelope there.

689 **Two-sided regret bound** (stated as Component 2’s two-sided characterization in Section 4.1 and as Key  
690 Lemma 1’s *Per-round regret bound* paragraph in Section 5.1). From the regret expression  $R(Q) =$   
691  $\sum_{a \neq a^*} Q(a) \Delta(a)$  and  $\Delta(a) \leq V_{\max}$ :  $R(Q) \leq V_{\max} \sum_{a \neq a^*} Q(a) = V_{\max}(1 - Q(a^*)) =$   
692  $V_{\max} \text{TV}(\delta_{a^*}, Q) = V_{\max}(1 - e^{-D_{\text{KL}}})$ . The matching action-gap lower bound (full statement in  
693 Appendix C.4) gives  $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \text{TV}(\delta_{a^*}, Q) =$   
694  $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$ . Combining gives the two-sided characterization. The bound is *Lipschitz-equivalent*  
695 with constants  $\Delta_{\min}/V_{\max}$  and 1, both achieved on extremal value landscapes — therefore *coordinate-*  
696 *optimal among bounds depending only on TV*.  $\square$

## 697 B.2 Proof of the perturbative extension — $\epsilon$ -stochastic and softmax-regularized

698 We establish the perturbative identity via a uniform perturbation argument rather than an alignment-  
699 specific calculation.

700  **$\epsilon$ -greedy stochastic optimum** Let  $\pi_{\epsilon}^*(a^*) = 1 - \epsilon$  and  $\pi_{\epsilon}^*(a) = \epsilon/(|\mathcal{A}| - 1)$  for  $a \neq a^*$ , and  
701 assume the full-support lower bound  $Q(a) \geq q_0 > 0$  for all  $a \in \mathcal{A}$ . (Full support is required since  $\pi_{\epsilon}^*$   
702 has positive mass on every action; without  $Q(a) > 0$  on the full support,  $D_{\text{KL}}(\pi_{\epsilon}^* \| Q) = +\infty$ .)

703 *Step 1 — TV is Lipschitz under the point-mass perturbation.*  $\text{TV}(\pi_{\epsilon}^*, \delta_{a^*}) = \epsilon$ . By the triangle  
704 inequality for total variation,  $|\text{TV}(\pi_{\epsilon}^*, Q) - \text{TV}(\delta_{a^*}, Q)| \leq \text{TV}(\pi_{\epsilon}^*, \delta_{a^*}) = \epsilon$ , uniformly in  $Q$  —  
705 no alignment hypothesis needed.

706 *Step 2 — Reverse-KL admits a uniform expansion.* Direct computation:

$$\begin{aligned} D_{\text{KL}}(\pi_{\epsilon}^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q) &= (1 - \epsilon) \log \frac{1 - \epsilon}{Q(a^*)} + \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log \frac{\epsilon/(|\mathcal{A}| - 1)}{Q(a)} - (-\log Q(a^*)) \\ &= (1 - \epsilon) \log(1 - \epsilon) + \epsilon \log Q(a^*) + \epsilon \log \frac{\epsilon}{|\mathcal{A}| - 1} - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a), \end{aligned}$$

708 where the second equality collects the  $\log Q(a^*)$  contributions via  $-(1 - \epsilon) \log Q(a^*) + \log Q(a^*) =$   
709  $\epsilon \log Q(a^*)$  and splits the constant  $\log$  term off the conditional sum. Expanding  
710  $(1 - \epsilon) \log(1 - \epsilon) = -\epsilon + O(\epsilon^2)$  and  $\epsilon \log(\epsilon/(|\mathcal{A}| - 1)) = \epsilon \log \epsilon - \epsilon \log(|\mathcal{A}| - 1)$ ,  
711 the leading-order behavior is  $D_{\text{KL}}(\pi_{\epsilon}^* \| Q) = D_{\text{KL}}(\delta_{a^*} \| Q) + \epsilon \log \epsilon - \epsilon(\log(|\mathcal{A}| - 1) + 1) + \epsilon \log Q(a^*) - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a) + O(\epsilon^2)$ . Under  $Q(a) \geq q_0$ , each  
712  $|\log Q(a)| \leq \log(1/q_0)$  is uniformly bounded; the dominant correction is the  $\epsilon \log \epsilon$  leading  
713 term, giving  $|D_{\text{KL}}(\pi_{\epsilon}^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q)| = O(\epsilon \log(1/\epsilon) + \epsilon \log(1/q_0)) = O(\epsilon \log(1/\epsilon))$   
714 absorbing  $\epsilon \log(1/q_0)$  into the leading  $\epsilon \log(1/\epsilon)$  term as  $\epsilon \rightarrow 0$  (constants depend on  $q_0$  and  $|\mathcal{A}|$ ).  
715

716 *Step 3 — Map through  $1 - e^{-x}$  via Lipschitzness.* The map  $x \mapsto 1 - e^{-x}$  is 1-Lipschitz on  $[0, \infty)$ ,  
717 so  $|(1 - e^{-D_{\text{KL}}(\pi_{\epsilon}^* \| Q)}) - (1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)})| = O(\epsilon \log(1/\epsilon))$ . Combining Steps 1 and 3 with  
718 the unperturbed identity  $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}$ :  $\text{TV}(\pi_{\epsilon}^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_{\epsilon}^* \| Q)} +$   
719  $O(\epsilon \log(1/\epsilon))$  uniformly in  $Q$  over the class  $\{Q : Q(a) \geq q_0\}$ , with constants depending on  $q_0$  and  
720  $|\mathcal{A}|$ . This establishes the  $\epsilon$ -greedy form of the perturbative identity.  $\square$

721 **Softmax-regularized optimum** Let  $\pi_{\tau}^*(a) \propto \exp(Q_O(a)/\tau)$ . Under isolated optimum with  
722 action gap  $\Delta_{\min}$ , the off-optimum mass concentrates exponentially in  $1/\tau$ :  $1 - \pi_{\tau}^*(a^*) =$   
723  $\frac{\sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)}{1 + \sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)} \leq (|\mathcal{A}| - 1) \exp(-\Delta_{\min}/\tau)$ . Treating  $\pi_{\tau}^*$  as  $\pi_{\epsilon}^*$  with ef-  
724 fective  $\epsilon = O(\exp(-\Delta_{\min}/\tau))$ , the  $\epsilon$ -greedy expansion above applies with  $\epsilon \log(1/\epsilon) =$   
725  $O((\Delta_{\min}/\tau) \exp(-\Delta_{\min}/\tau)) = O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ , exponentially small with a polynomial  
726 prefactor (equivalently,  $O(\exp(-c\Delta_{\min}/\tau))$  for any fixed  $c \in (0, 1)$  since the exponential dominates  
727 the polynomial).  $\square$



728 **Two-sided regret bound under perturbation** Composing with the TV-regret bounds:  $\Delta_{\min}(1 -$   
729  $e^{-D_{\text{KL}}(\pi_{\epsilon}^* \parallel Q)}) - O(\epsilon \log(1/\epsilon)) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi_{\epsilon}^* \parallel Q)}) + O(\epsilon \log(1/\epsilon)).$

### 730 B.3 Proof of Key Lemma 2 (Multi-factor forgetting prerequisite)

731 We work in the continuous-time idealization of Model  $(\Sigma)$ :  $\dot{\delta}_{\Sigma} = -\mathbf{F}(\delta_{\Sigma}) + \mathbf{w}$  with  $\mathbf{F}(\delta_{\Sigma}) :=$   
732  $(\alpha_{ij}^{\text{ss}} \delta_{ij})_{(i,j) \in E}$  the sector correction and  $\mathbf{w}$  the deterministic disturbance with  $\|\mathbf{w}\| \leq \rho_{\Sigma}$ . The  
733 discrete-time form recovers the same scaling up to absorbable constants via an AM–GM expansion of  
734  $\|\delta_{\Sigma} - \mathbf{F} + \mathbf{w}\|^2$  (sketched below).

735 For the Lyapunov function  $V(\delta_{\Sigma}) := \|\delta_{\Sigma}\|^2$ ,  $\dot{V} = 2\delta_{\Sigma}^{\top} \dot{\delta}_{\Sigma} = -2\delta_{\Sigma}^{\top} \mathbf{F}(\delta_{\Sigma}) + 2\delta_{\Sigma}^{\top} \mathbf{w}$ . The  
736 first term is the *sector product*, bounded below by the bottleneck:  $\delta_{\Sigma}^{\top} \mathbf{F}(\delta_{\Sigma}) = \sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2 \geq$   
737  $\min_{(i,j)} \alpha_{ij}^{\text{ss}} \cdot \|\delta_{\Sigma}\|^2 = \mathcal{T}_{\Sigma}^{\text{bn,ss}} \cdot \|\delta_{\Sigma}\|^2$ , with the bound tight: setting  $\delta_{\Sigma}$  proportional to  $\mathbf{e}_{(i^*, j^*)}$  where  
738  $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$  saturates it. The cross-term is bounded by Cauchy–Schwarz:  $|2\delta_{\Sigma}^{\top} \mathbf{w}| \leq$   
739  $2\|\delta_{\Sigma}\| \cdot \|\mathbf{w}\| \leq 2\rho_{\Sigma} \|\delta_{\Sigma}\|$ . Combining:  $\dot{V} \leq -2\mathcal{T}_{\Sigma}^{\text{bn,ss}} \|\delta_{\Sigma}\|^2 + 2\rho_{\Sigma} \|\delta_{\Sigma}\| = -2\|\delta_{\Sigma}\| \cdot$   
740  $(\mathcal{T}_{\Sigma}^{\text{bn,ss}} \|\delta_{\Sigma}\| - \rho_{\Sigma})$ . The drift is strictly negative whenever  $\|\delta_{\Sigma}\| > \rho_{\Sigma}/\mathcal{T}_{\Sigma}^{\text{bn,ss}}$ ; the trajectory is  
741 ultimately bounded by  $R_{\Sigma}^* = \rho_{\Sigma}/\mathcal{T}_{\Sigma}^{\text{bn,ss}}$ . So whenever  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$ , we have  $R_{\Sigma}^* \leq R_{\Sigma}$  and  
742 the modeled mismatch is ultimately bounded within the strategic reserve. This is the standard  
743 Khalil sector-Lyapunov ultimate-boundedness setup with the cross-term retained<sup>[46]</sup> (Section 9.2;  
744 comparison-principle form).  $\square$

745 *Discrete-time form.* For the discrete update  $\delta_{\Sigma}^+ = \delta_{\Sigma} - \mathbf{F}(\delta_{\Sigma}) + \mathbf{w}$  with  $\alpha_{ij}^{\text{ss}} \in [0, 1]$  and AM–GM  
746 with parameter  $\eta = \mathcal{T}_{\Sigma}^{\text{bn,ss}}/(2(1 - \mathcal{T}_{\Sigma}^{\text{bn,ss}}))$ , the steady-state bound is  $\|\delta_{\Sigma}^*\| \leq 2\rho_{\Sigma}/\mathcal{T}_{\Sigma}^{\text{bn,ss}}$  — same  
747 scaling, factor-of-2 absorbable into the threshold constant.

748 *Mean-square corollary.* If instead  $\mathbf{w}$  is interpreted as zero-mean stochastic disturbance with  $\mathbb{E}[\|\mathbf{w}\|^2] \leq$   
749  $\rho_{\Sigma}^2$ , the cross-term  $\mathbb{E}[2\delta_{\Sigma}^{\top} \mathbf{w} \mid \delta_{\Sigma}] = 0$  vanishes in expectation, and the drift inequality gives mean-  
750 square ultimate boundedness  $\sqrt{\mathbb{E}[V^*]} \leq \rho_{\Sigma}/\sqrt{2\mathcal{T}_{\Sigma}^{\text{bn,ss}}}$  under threshold  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}^2/(2R_{\Sigma}^2)$ . The  
751 deterministic-disturbance form above is the operative one for Theorem 4.1(A2) since non-stationarity-  
752 induced drift is structurally adversarial-aligned (consistently moving mismatch in one direction) rather  
753 than zero-mean.

754 **Sharpness** When the inequality reverses ( $\mathcal{T}_{\Sigma}^{\text{bn,ss}} \leq \rho_{\Sigma}/R_{\Sigma}$ ), the adversarial disturbance  $\mathbf{w}^*(t) =$   
755  $\rho_{\Sigma} \delta_{\Sigma}(t)/\|\delta_{\Sigma}(t)\|$  saturates Cauchy–Schwarz and gives  $d\|\delta_{\Sigma}\|/dt \geq -\mathcal{T}_{\Sigma}^{\text{bn,ss}} \|\delta_{\Sigma}\| + \rho_{\Sigma}$ . By the  
756 comparison principle,  $\|\delta_{\Sigma}(t)\|$  exceeds any  $R < \rho_{\Sigma}/\mathcal{T}_{\Sigma}^{\text{bn,ss}}$  in finite time. The bottleneck-element  
757 refinement: pick  $\mathbf{w}$  and  $\delta_{\Sigma}$  both supported on  $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$  with  $|w_{i^* j^*}| = \rho_{\Sigma}$  —  
758 saturates the sector inequality on the contracting side and aligns with  $\mathbf{w}$  on the disturbance side  
759 simultaneously. The threshold is *sharp inside the diagonal sector model*. The theorem is silent about  
760 non-diagonal correction architectures or stabilization mechanisms outside Model  $(\Sigma)$ .  $\square$

### 761 B.4 Proof of the impulsive ProST sector-level reduction

762 Within Model  $(\Sigma)$ , idealize ProST<sup>[8]</sup> as an impulsive system: between scheduled update times  
763  $\{t_1, \dots, t_K\} \subset [0, T]$  the agent’s policy is held fixed and the strategic mismatch evolves under  
764 disturbance budget  $\rho_{\Sigma}$  alone (continuous-destabilizing in the Lyapunov  $V = \|\delta_{\Sigma}\|^2$ ); at each update  
765 time  $t_i$  the policy is revised, contracting the modeled mismatch by per-update impulse gain  $\gamma \in (0, 1]$   
766 so  $\|\delta_{\Sigma}(t_i^+)\| \leq (1 - \gamma)\|\delta_{\Sigma}(t_i^-)\|$ . Then  $V(t_i^+) \leq (1 - \gamma)^2 V(t_i^-)$ .

767 This fits the Hespanha–Liberzon–Teel<sup>[44]</sup> (Theorem 1) framework for impulsive systems with desta-  
768 bilizing continuous evolution and stabilizing impulses (the *reverse-ADT* branch). The continuous-  
769 evolution rate is  $\lambda_c = \rho_{\Sigma}/R_{\Sigma}$  (from linearizing the disturbance contribution to  $\dot{V} \approx +\lambda_c V + \gamma_c(\rho_{\Sigma})$ ).  
770 The impulse contraction in  $V$  is  $\mu = (1 - \gamma)^2$ . The reverse-ADT condition for ultimate boundedness —  
771 impulses cannot be too sparse — gives  $\Delta_{\max} \cdot \lambda_c < -\ln \mu \iff \Delta_{\max} \cdot \rho_{\Sigma}/R_{\Sigma} < -\ln(1 - \gamma)^2$ ,  
772 where  $\Delta_{\max} := \sup_i (t_{i+1} - t_i)$  is the longest inter-update gap. Under the condition the modeled  
773 mismatch is ultimately bounded within  $R_{\Sigma}^* \approx \rho_{\Sigma} \Delta_{\max}/\gamma$  (leading order in  $\gamma$ ).

Under uniform blocks  $\Delta_i = T/K$  the condition reduces to  $K/T > \rho_\Sigma / (2\gamma R_\Sigma)$  to leading order in  $\gamma$  — recovering ProST’s  $K/T$  form with the impulse gain made explicit. Under nonuniform schedules the impulsive form is *strictly stronger*: the longest block sets the threshold, not the average. The lemma is rigorous for the modeled impulsive sector dynamics; the per-update impulse gain  $\gamma$  is a domain quantity (depends on within-block sample size and update-time discount) and is left abstract.  $\square$

## B.5 Proof of Key Lemma 3 (Loop generates interventional samples)

By temporal ordering and the singular-trajectory commitment of Section 3,  $a_t$  causally precedes  $o_{t+1}$ . We invoke Pearl’s do-calculus<sup>[26]</sup> (Theorem 3.4.1, Rule 2 — *action/observation exchange*):  $P(o_{t+1} \mid \text{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$  whenever  $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\overline{a_t}}}$ , where  $G_{\overline{a_t}}$  denotes the graph  $G$  with all incoming arrows to  $a_t$  removed. In the singular-trajectory + agent-as-policy graph of Section 3, the only arrows into  $a_t$  come from  $H_t$  (the agent’s policy is  $\pi_t(a_t \mid H_t)$ ); deleting them severs the only confounding path from  $a_t$  to  $o_{t+1}$ . Whether  $H_t$  d-separates  $a_t$  from  $o_{t+1}$  in  $G_{\overline{a_t}}$  is precisely the content of (C2). The identity  $P(o_{t+1} \mid \text{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$  holds on the support where  $\pi_t(a_t \mid H_t) > 0$ , which is automatic since  $a_t$  was drawn from  $\pi_t$ .  $\square$

*Remark on equivalent forms of (C2).* The d-separation condition  $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\overline{a_t}}}$  is equivalent to the potential-outcome form  $a_t \perp Y^{(a_t)} \mid H_t$  (sequential ignorability in the Robins/Murphy sense), and to the truncated-factorization condition  $P(\mathbf{U} \mid a_t, H_t) = P(\mathbf{U} \mid H_t)$  where  $\mathbf{U}$  are the latent variables jointly affecting  $a_t$  and  $o_{t+1}$  — applied to the *prior* on  $\mathbf{U}$  in the do-marginalization, not to the conditional on  $o_{t+1}$ . The three forms are interchangeable; the d-separation form is the most direct handle for the proof above.

*On the role of (C1) and (C3).* The identification proof above is load-bearing on (C2) alone: (C1)’s full-support positivity and (C3)’s known-action-mechanism are *automatic* under the paper’s architecture — (C1) reduces to realized-action positivity ( $\pi_t(a_t \mid H_t) > 0$  tautologically since  $a_t$  was drawn from  $\pi_t$ ); (C3) reduces to agent-policy-queryability ( $Q_t$  is the agent’s own policy, internally accessible). Both are stated as named conditions for forward-compatibility with off-policy / counterfactual extensions where they would do real work; for the on-policy identification claim of this lemma, (C2) is the substantive condition.

## B.6 Proof of Key Lemma 4 (Bias bound)

On the matching event  $\{a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)\}$ , the difference  $\hat{D}_t(s) - D_t^{\text{true}}(s)$  vanishes identically (both quantities equal  $-\log Q_t$  at the same action). On the complementary misidentification event, the support condition  $Q_t(\cdot \mid s) \geq q_0$  at *both*  $a_{\text{ag},t}^*(s)$  and  $\tilde{a}_t^*(s)$  implies  $\log Q_t(a_{\text{ag},t}^*(s) \mid s) \in [\log q_0, 0]$  and  $\log Q_t(\tilde{a}_t^*(s) \mid s) \in [\log q_0, 0]$ , so their absolute difference is at most  $\log(1/q_0)$ . Therefore  $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$ . Taking expectations:  $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$  by definition of  $p_{\text{id}}(s)$ .  $\square$

**High-probability form** With probability  $1 - \delta$  over a single round’s identification event,  $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \log(1/q_0) \cdot \mathbf{1}[\text{misdevent}]$  where  $\Pr[\text{misdevent}] = 1 - p_{\text{id}}(s)$ . The bias decomposes cleanly: in Regime A ( $p_{\text{id}}(s) \rightarrow 1$ ) it vanishes; in Regime B ( $p_{\text{id}}(s) \in (0, 1)$ ) it is controlled by misidentification mass; in Regime C ( $p_{\text{id}}(s) \rightarrow 0$ ) it saturates at  $\log(1/q_0)$ . This proves Key Lemma 4.  $\square$

**Per-state, per-horizon-step lift (used in Theorem 4.1(v)).** The lemma applies pointwise at any visited state  $s_h$  along a horizon- $N_h$  trajectory:  $\mathbb{E}[|\hat{D}_t(s_h) - D_t^{\text{true}}(s_h)|] \leq (1 - p_{\text{id}}(s_h)) \log(1/q_0)$ , with  $p_{\text{id}}(s_h)$  the per-state argmax-correctness probability. Aggregating linearly over the horizon and assuming the per-state floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ , the per-round bias contribution to the trajectory-level value gap is at most  $N_h(1 - p_{\text{id}}) \log(1/q_0)$ , contributing the second term of Theorem 4.1(v).

**Remarks** - Both  $a_{\text{ag},t}^*(s)$  and  $\tilde{a}_t^*(s)$  must satisfy  $Q_t \geq q_0$  for the  $\log(1/q_0)$  ceiling to apply; the support condition is on *both* points, not just the agent’s estimate. This is *strictly weaker* than the perturbative identity’s full-support condition  $Q(a) \geq q_0$  for *all*  $a$  (Appendix B.2) — the perturbative identity needs full support because  $\pi_e^*$  has positive mass on every action; the bias bound here needs only two-point support. - This is *not* an estimation-from-samples lemma. If  $Q_t$  is unavailable internally

and must be estimated from on-policy rollouts, an additional concentration argument (Hoeffding under iid fixed-policy samples, or martingale concentration for adaptive policies) is needed; in the architectures this paper considers,  $Q_t$  is the agent’s policy and is internally available.

## C Auxiliary Mathematical Material

This appendix collects supporting material used by the main results: convention hierarchy details, the admissible-divergence family, direction-forcing argument, action-gap matching lower bound, tied-optimum extensions, the structural-class theorem on gain-decay updates, strategic-tempo topologies, deployment modes, and the chain-rule uniqueness theorem.

### C.1 Convention hierarchy: C1, C2, C3

Three named continuation conventions form a hierarchy of increasing diagnostic power and computational cost.

**C1 — One-step improvement (canonical default)**  $\pi_{\text{cont}} = \pi_{\text{current}}$ . Each action is evaluated assuming current behavior continues afterward. No fixed-point computation, no global-optimality assumption. Cheapest; weakest diagnostic.

**C2 — Receding-horizon ( $N_r$ -step replanning)** At each future step, re-optimize over a horizon of  $N_r$  steps using the model available at that step:  $\pi_{\text{RH}}(M_\tau) = \arg \max_\pi V_O(M_\tau, \pi; N_r)$  applied at each  $\tau$ . Captures multi-step recovery: a goal that appears unattainable under frozen continuation may be reachable with replanning. Moderate cost; moderate diagnostic.

**C3 — Bellman**  $\pi_{\text{cont}} = \pi^*$  where  $\pi^* = \arg \max_\pi V_O(M_t, \pi; N_h)$ . The continuation is the optimal policy — a fixed-point equation. Strongest diagnostic; most expensive.

**Proposition C.1 (Monotonicity).** *For any model  $M_t$ , horizon  $N_h$ , and policy class  $\Pi$ :  $A_O^{(1)}(M_t; \Pi, N_h) \leq A_O^{\text{RH}}(M_t; \Pi, N_r, N_h) \leq A_O^{\text{B}}(M_t; \Pi, N_h)$ .*

*Proof.* C1 freezes continuation at  $\pi_{\text{current}}$  (possibly suboptimal); C2 re-optimizes periodically, so  $\pi_{\text{RH}} \succeq \pi_{\text{current}}$  at each future step; C3 uses the globally optimal  $\pi^*$ . A weakly better continuation yields a weakly higher expected trajectory value; taking the supremum over the first action preserves the ordering.  $\square$

**Corollary**  $\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)}$  (since  $\delta_{\text{sat}} = V_O^{\min} - A_O$ , higher  $A_O$  means lower  $\delta_{\text{sat}}$ ). And  $\delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}}$  (since  $\delta_{\text{regret}} = A_O - V_O(M_t, \pi_{\text{current}}; N_h)$ , higher  $A_O$  means higher  $\delta_{\text{regret}}$ ).

C1 is the most conservative diagnostic (most likely to diagnose “locally unattainable”); C3 is the most accurate (least false “unattainable” diagnoses). The  $2 \times 2$  diagnostic structure is preserved under all three; only the inferential force varies.

### C.2 Admissible-divergence family for the regret bound

The reverse-KL direction is forced by the deterministic- $\pi^*$  regime (Appendix C.3). Within the direction-forced family, multiple smooth  $f$ -divergences yield valid regret bounds:

| Divergence                                          | Bound on $R(Q)$                            | Tightness               | Finite under det. $\pi^*$ ? |
|-----------------------------------------------------|--------------------------------------------|-------------------------|-----------------------------|
| $\text{TV}(\pi^*, Q)$                               | $V_{\max} \cdot \text{TV}$                 | Tight (extremal $V$ )   | Yes                         |
| $D_{\text{KL}}(\pi^* \  Q)$ via Pinsker             | $V_{\max} \sqrt{D_{\text{KL}}/2}$          | Loose by $\sqrt{\cdot}$ | Yes                         |
| $D_{\text{KL}}(\pi^* \  Q)$ via point-mass identity | $V_{\max}(1 - e^{-D_{\text{KL}}})$         | Tight (this paper)      | Yes                         |
| $\chi^2(\pi^* \  Q)$ (Le Cam)                       | $V_{\max} \cdot \frac{1}{2} \sqrt{\chi^2}$ | Looser than Pinsker     | $\chi^2 = 1/Q(a^*) - 1$     |
| $D_\alpha(\pi^* \  Q)$ (Rényi, $\alpha \geq 1$ )    | Various                                    | Interpolates            | Yes for $\alpha \geq 1$     |
| $D_{\text{KL}}(Q \  \pi^*)$ (forward-KL)            | $+\infty$                                  | Vacuous                 | No                          |

Reverse-KL is selected uniquely within the direction-forced family by the chain-rule axiom (Appendix C.10). The point-mass identity supplies the *exact* bound on reverse-KL under deterministic  $\pi^*$ ; the table reflects different bound shapes on the same divergence (with the BH inequality  $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$  as the looser general form to which the identity reduces outside the deterministic- $\pi^*$  corner).

### C.3 Direction-forcing argument

For deterministic  $\pi^* = \delta_{a^*}$  and any  $Q$  with  $Q(a) > 0$  for some  $a \neq a^*$ :  $D_{\text{KL}}(Q \| \pi^*) = \sum_a Q(a) \log \frac{Q(a)}{\pi^*(a)} = \sum_{a \neq a^*} Q(a) \log \frac{Q(a)}{0} = +\infty$ . A bound “ $R \leq +\infty$ ” is vacuous. The reverse direction  $D_{\text{KL}}(\pi^* \| Q)$  is finite (and equal to  $-\log Q(a^*)$ ) whenever  $Q(a^*) > 0$ . The asymmetry is forced by the regime: regret is a one-sided quantity (contributes only from  $Q$ ’s off-optimum mass;  $\pi^*$  has no support off  $a^*$ ); divergences whose role is to bound this one-sided quantity must themselves be one-sided. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized KL) introduce a vacuous symmetric term.

### C.4 Action-gap matching lower bound

For any  $Q$  with  $\Delta_{\min} = \min_{a \neq a^*} \Delta(a) > 0$ :  $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \cdot (1 - Q(a^*)) = \Delta_{\min} \cdot \text{TV}(\pi^*, Q)$ . By the point-mass identity (Lemma 5.1),  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$ , giving the matching lower bound used in the two-sided characterization.

The lower bound is tight when sub-optimal actions are uniformly bad ( $\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$ ). For typical landscapes the gap between upper and lower bound is  $V_{\max} - \Delta_{\min}$ , controlled by the *spread* of action gaps.

### C.5 Tied-optimum and softmax-smoothed extensions

**Tied-optimum** If  $\pi^*$  has support on a tied-optimum set  $\mathcal{A}^* = \{a : Q_O(a) = Q_O(a^*)\}$  with uniform mass, reverse-KL is finite whenever  $Q$  covers  $\mathcal{A}^*$ . The regret bound extends:  $R(Q) = \sum_{a \notin \mathcal{A}^*} Q(a) \Delta(a) \leq V_{\max} \cdot \mathbb{P}_Q(a \notin \mathcal{A}^*)$ . A multi-action analog of the point-mass identity holds with  $\pi^*(a^*) = 1/|\mathcal{A}^*|$  replacing the point-mass form, yielding  $D_{\text{KL}}(\pi^* \| Q) = -\sum_{a \in \mathcal{A}^*} (1/|\mathcal{A}^*|) \log(|\mathcal{A}^*| Q(a))$ , a multi-action analog of  $-\log Q(a^*)$ . The two-sided bound becomes one-sided in general; the upper bound holds with the modified KL form.

**Softmax-smoothed  $\pi^*$  with temperature  $\tau \rightarrow 0$**  Replace  $\pi^* = \delta_{a^*}$  with  $\pi_\tau^* \propto \exp(Q_O/\tau)$ . As  $\tau \rightarrow 0$ ,  $\pi_\tau^* \rightarrow \delta_{a^*}$ . For finite  $\tau$ , the *perturbative* identity (Appendix B.2) applies:  $\text{TV}(\pi_\tau^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\tau^* \| Q)} + O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$  — exponentially small with polynomial prefactor. The two-sided regret bound transfers with the same correction order. Outside the perturbative regime — for genuinely high-entropy optima — the BH inequality  $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$  becomes the relevant general bound; Pinsker also applies, with a one-sided regret form.

### C.6 The gain-decay structural class $\mathcal{A}_{\text{decay}}$

The persistence inequality  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$  (Section 5.2) is an *instantaneous* check at the current operating point. Define the structural class by gain-decay rather than sample-counting:  $\mathcal{A}_{\text{decay}} := \{\text{updates whose effective sector gain } \alpha_{ij}^{(t)} \rightarrow 0 \text{ as } t \rightarrow \infty \text{ on every revisable element}\}$ .

**Theorem C.2** (Universal failure of  $\mathcal{A}_{\text{decay}}$ ). *For any fixed  $(\rho_{\Sigma}, R_{\Sigma})$  with  $\rho_{\Sigma} > 0$ , every agent in  $\mathcal{A}_{\text{decay}}$  eventually violates the persistence threshold  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$ .*

*Proof.* At every element  $(i, j)$ ,  $\alpha_{ij}^{(t)} \rightarrow 0$  by definition of  $\mathcal{A}_{\text{decay}}$ , so  $\min_{(i,j)} \alpha_{ij}^{(t)} \rightarrow 0$ , so the bottleneck  $\mathcal{T}_{\Sigma}^{\text{bn}} \rightarrow 0$  with experience. Once  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} < \rho_{\Sigma}/R_{\Sigma}$ , the threshold is violated. This holds *regardless of prior calibration*: at any finite calibration level, the gain decays below threshold once enough experience accumulates.  $\square$

903 *Scope clarification (per-element pull).* The class definition requires  $\alpha_{ij}^{(t)} \rightarrow 0$  on *every* revisable  
 904 element — i.e., the agent is being pulled at every element repeatedly. Beta-Bernoulli with  $\eta_{\text{edge},ij} =$   
 905  $1/(n_{ij} + 1)$  is in the class only for elements actually being updated; an agent that ignores a subset of  
 906 elements is outside the class for those elements (their gain remains at initial calibration), but inside  
 907 it for the elements it does touch. The theorem applies to the touched elements; persistence on the  
 908 untouched ones reduces to whatever calibration was set initially.

909 This is a *structural* failure of the class, not a tuning problem. The class includes: - Count-accumulating  
 910 Bayesian updates without forgetting (e.g., Beta-Bernoulli with  $\eta_{\text{edge},ij} = 1/(n_{ij} + 1)$  where  $n_{ij} \rightarrow \infty$ ).  
 911 - Bounded-memory schemes with growing memory. - Observation-aggregating schemes without restart.  
 912 - Gradient-based methods with vanishing step sizes ( $\eta_t = c/t^\beta$  for  $\beta \in (0, 1]$ , the Robbins–Monro  
 913 regime).

914 Mechanisms *outside*  $\mathcal{A}_{\text{decay}}$  — constant-step-size stochastic approximation, sliding-window updates,  
 915 bounded-memory learners, block-restart schemes, Kalman filters with bounded process noise —  
 916 maintain a finite gain ceiling and escape asymptotic decay, but then face a *bidirectional* threshold:

Table 1: Bidirectional thresholds for non-accumulating mechanisms outside  $\mathcal{A}_{\text{decay}}$ .

| Mechanism                                                                                                 | $\alpha_\Sigma^{\text{ss}}$ | Prerequisite holds iff                          |
|-----------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------|
| Exponential forgetting with $\lambda$                                                                     | $1 - \lambda$               | $1 - \lambda > \rho_\Sigma / R_\Sigma$          |
| Constant-step SA with rate $\eta$                                                                         | $\eta$                      | $\eta > \rho_\Sigma / R_\Sigma$                 |
| Fixed-window mechanisms (sliding-window size $W$ , bounded-memory size $K$ , block-restart period $T_R$ ) | $1/W, 1/K, 1/T_R$           | window/memory/period $< R_\Sigma / \rho_\Sigma$ |

917 For each row, the threshold is *bidirectional* and sharp within the sector-Lyapunov reduction.

## 918 C.7 Strategic-tempo across canonical topologies

919 The bottleneck strategic tempo  $\mathcal{T}_\Sigma^{\text{bn,ss}}$  of Lemma 5.2 is verified — and the resulting per-topology  
 920 threshold derived — across four canonical topologies under Beta-Bernoulli edge updates with per-  
 921 element forgetting:

- 922 • **(T1) Single edge.**  $\mathcal{T}_\Sigma^{\text{bn,ss}} = \nu \cdot \iota \cdot (1 - \lambda)$ . All three factors load-bearing; setting any one to  
 923 zero collapses the bottleneck.
- 924 • **(T2) Two-edge AND chain, observable intermediate.**  $\mathcal{T}_\Sigma^{\text{bn,ss}} = \min(\nu_1(1 - \lambda_1), \nu_1\theta_1(1 -$   
 925  $\lambda_2))$ . Edge 2’s effective observation rate is gated by edge 1’s success probability  $\theta_1$  —  
 926 *depth-gated attenuation*. For depth- $d$  chains, the bottleneck is  $\min_k \prod_{j < k} \theta_j(1 - \lambda_k)$ .
- 927 • **(T3) Two-edge AND chain, unobservable intermediate.** Per-edge tempo ill-defined; plan-  
 928 level bottleneck  $\mathcal{T}_{\Sigma, \text{plan}}^{\text{bn,ss}} = \nu(1 - \lambda_\Phi)$  over a single tracked plan-level quantity  $\hat{\Phi} = p_1 p_2$ .
- 929 • **(T4) Two-arm OR node,  $\varepsilon$ -greedy.**  $\mathcal{T}_\Sigma^{\text{bn,ss}} = \min((1 - \varepsilon)(1 - \lambda_1), \varepsilon(1 - \lambda_2))$ . Action  
 930 selection controls rate allocation; pure greedy ( $\varepsilon = 0$ ) collapses the unexplored arm’s  
 931 bottleneck — *exploration-gated*.

932 The structural decomposition: AND-chains exhibit *depth-gated* geometric attenuation  $\nu_k =$   
 933  $\nu \prod_{j < k} \theta_j$ ; OR-nodes exhibit *exploration-gated* allocation. Mixed AND/OR DAGs interleave both.  
 934 Each topology surfaces a different factor as the binding bottleneck.

935 **Regime-C contamination is structurally fatal**  $\iota_{ij} = 0$  at any single element  $\Rightarrow \mathcal{T}_\Sigma^{\text{bn,ss}} = 0$ . An  
 936 agent with even one fully-confounded element cannot meet the threshold by forgetting alone.

## 937 C.8 Three deployment modes of loop-Level-2

938 The Pearl-do structure of closed-loop interventional access manifests at distinct layers with semanti-  
 939 cally distinct interventional mechanisms:

- **Mode 1 — agent-self-intervention.** The agent performs do-actions on its own action space as part of its ordinary adaptive loop. Intervention is on the agent’s own action; target is the environment’s response. This is Lemma 5.3’s primary content.
- **Mode 2 — observer-on-sub-agent.** An observer external to a composite performs do-interventions on one sub-agent’s action space; target is another sub-agent’s response. Reveals cross-coupling structure that component-marginal observation cannot identify.
- **Mode 3 — observer-on-agent-input** (sketched, future work). An observer intervenes on the agent’s observation channel; target is the agent’s subsequent policy. Useful for offline architectural-class diagnosis.

Modes share the Pearl-do structure but differ in who intervenes on what. The unification is at the pattern level; the mechanism is semantically distinct per layer.

## C.9 Distinction from active inference and causal-RL precursors

Action-perception-loop frameworks — active inference<sup>[47,48]</sup>, control-as-inference<sup>[49]</sup>, cybernetics<sup>[50,51]</sup> — implicitly use the action-causes-observation observation. Our distinctive moves:

- *Bareinboim-hierarchy connection.* Active inference / cybernetics rest on Bayesian-network (Level 1) generative models; we invoke the causal-hierarchy theorem<sup>[27]</sup> to position the policy DAG as causal with do-conditioning in  $Q_O$ .
- *Regime-indexed identifiability (A/B/C).* Active inference literature treats causal identifiability uniformly within modeling assumptions; we surface the regime split at framework level.
- *Scope honesty.* We distinguish “data generated under intervention” from “cleanly identified do-estimates”;<sup>[45]</sup> documents that the active-inference literature sometimes elides this.

The causal-RL line<sup>[13–18]</sup> is the direct ancestor for regime-indexed identifiability and on-policy interventional access; all are stationary-MDP. Composition with non-stationarity is, to our knowledge, novel.

## C.10 Chain-rule uniqueness of reverse-KL

**Theorem C.3** (Chain-rule uniqueness (Hobson 1969; Csiszár 1991)). *Let  $D_f(P \parallel Q) = \sum_x Q(x) f(P(x)/Q(x))$  be a smooth  $f$ -divergence with  $f$  convex and  $f(1) = 0$ . The chain rule  $D_f(P_{XY} \parallel Q_{XY}) = D_f(P_X \parallel Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \parallel Q_{Y|X})]$  holds for all joint distributions if and only if  $f(t) = c \cdot t \log t$  for some  $c > 0$  — i.e.,  $D_f$  is reverse-KL up to positive scaling.*

*Proof.* Take any joint  $P_{XY}, Q_{XY}$  on a finite product space. Writing  $r_x := P(x)/Q(x)$  and  $s_{y|x} := P(y|x)/Q(y|x)$ , the joint ratio factorizes as  $P_{XY}(x, y)/Q_{XY}(x, y) = r_x \cdot s_{y|x}$ . Expanding both sides of the chain rule:  $D_f(P_{XY} \parallel Q_{XY}) = \sum_{x,y} Q_{XY}(x, y) f(r_x s_{y|x}) = \sum_x Q(x) \sum_y Q(y|x) f(r_x s_{y|x})$ ,  $D_f(P_X \parallel Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \parallel Q_{Y|X})] = \sum_x Q(x) f(r_x) + \sum_x P(x) \sum_y Q(y|x) f(s_{y|x})$ , the second term using  $\mathbb{E}_{P_X} = \sum_x P(x) = \sum_x Q(x) r_x$ . Equating and collecting:  $\sum_x Q(x) \left[ \sum_y Q(y|x) f(r_x s_{y|x}) - f(r_x) - r_x \sum_y Q(y|x) f(s_{y|x}) \right] = 0$ . The chain rule must hold for *all* joints, including those where  $r_x$  takes arbitrary positive value at one  $x$  and  $s_{y|x}$  varies independently — which forces the bracketed expression to vanish pointwise. A two-point reduction (take  $Y$  binary with  $s_{y|x}$  taking values  $s$  and  $1/s$  on equal-weight  $Q(y|x)$ ) collapses the inner sums and yields the functional equation  $f(rs) = f(r) + r f(s)$  for all  $r, s > 0$ . With  $f$  convex and  $f(1) = 0$ , the unique solution is  $f(t) = c \cdot t \log t$  for  $c > 0$ <sup>[52]</sup> (§4) — this gives  $D_f(P \parallel Q) = c \sum_x Q(x) (P(x)/Q(x)) \log(P(x)/Q(x)) = c \sum_x P(x) \log(P(x)/Q(x)) = c \cdot D_{\text{KL}}(P \parallel Q)$ , reverse-KL up to positive scaling.  $\square$

**References** Hobson 1969 (“A new theorem of information theory,” *J. Stat. Phys.*); Csiszár 1991 (“Why least squares and maximum entropy?” *Annals of Statistics*; Theorem 3 corollary, Theorem 5); Shore-Johnson 1980 (“Axiomatic derivation of the principle of maximum entropy,” *IEEE Trans. Info. Theory*, system-independence axiom); Sanov 1957 (large-deviation rate function); Aczél-Daróczy 1975 (functional-equation machinery).

987 These references give *structurally equivalent reformulations* of the same axiom. The Cauchy functional  
 988 equation each reduces to is the common content. No known uniqueness route outside the independence-  
 989 on-sub-problems family exists.

990 **Why other family members fail the chain rule** Concrete counterexample for  $\chi^2$ : take  $Q_X$  uniform  
 991 on  $\{x_1, x_2\}$ ,  $P_X = (3/4, 1/4)$ ,  $Q(y|x)$  uniform,  $P(y|x) = (3/4, 1/4)$ . Direct calculation gives  
 992  $\chi^2(P_{XY} \| Q_{XY}) = 9/16$  while  $\chi^2(P_X \| Q_X) + \mathbb{E}_{P_X}[\chi^2(P_{Y|X} \| Q_{Y|X})] = 1/4 + 1/4 = 8/16$ .  
 993 Non-additive. Rényi- $\alpha$  for  $\alpha \neq 1$  fails analogously; squared Hellinger likewise fails.

## 994 D Algorithm Sketch and Base-Learner Instantiation

995 A practical algorithm achieving the joint guarantees of Theorem 4.1 requires four components, each  
 996 translating one of the assumptions (A1)–(A5) into operational form.

997 **(a) Identity-tight base learner satisfying (A5)** In the stationary deterministic- $\pi^*$  bandit case ( $N_h =$   
 998  $1$ ), Thompson sampling<sup>[19]</sup> (posterior-randomized; pointwise  $Q_t(a^*) > 0$ ) and vanilla deterministic  
 999 UCB<sup>[34]</sup> (Dirac-deployed;  $Q_t(a^*) \in \{0, 1\}$ , with the (A1) extended-real reading of Theorem 4.1) both  
 1000 achieve the per-round coordinate in expectation:  $\mathbb{E}[1 - e^{-K_t}] = \mathbb{E}[1 - Q_t(a^*)] = O\left(\frac{\log t}{t \Delta_{\min}^2}\right)$ ,  
 1001 following from Lattimore and Szepesvári 2020<sup>[34]</sup> Theorem 7.1 ( $\mathbb{E}[N_a(t)] \leq 16 \log(t)/\Delta_a^2 + O(1)$ )  
 1002 per suboptimal arm; aggregating over arms with the per-state floor gives the per-round form). This  
 1003 satisfies (A5) of Theorem 4.1 with the *stronger logarithmic* rate, yielding cumulative dynamic  
 1004 regret  $O(V_{\max}(B_T + 1) \log(T/(B_T + 1))/\Delta_{\min}^2)$ . The  $V_{\max} \cdot \text{TV}$  chain is structurally one factor of  
 1005  $\Delta_{\min}$  looser than direct gap-aware UCB analysis (which gives  $O(\log T/\Delta_{\min})$  cumulative regret per  
 1006 block by weighting each suboptimal pull by its own  $\Delta(a)$  rather than  $V_{\max}$ ); this is a feature of the  
 1007 unification — the framework’s contribution is composition + non-stationarity-handling + identity-  
 1008 coordinate exactness, not a sharper bandit constant. For  $N_h > 1$  MDPs, UCRL2<sup>[53]</sup> and UCBVI<sup>[41]</sup>  
 1009 give per-block cumulative trajectory-level regret  $\tilde{O}(N_h^{3/2} \sqrt{SAL})$  with  $S, A$  the state/action sizes;  
 1010 rewriting in the occupancy-weighted coordinate satisfies (A5) with  $c$  of order  $\tilde{O}(\sqrt{N_h SA})$ , lifting  
 1011 cleanly to the  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  cumulative rate — same piecewise-stationary  $B_T$  family as  
 1012 restart-on-change variants of<sup>[1]</sup> (the continuous-variation regime of<sup>[3]</sup> is structurally distinct; see  
 1013 Section 4 *Comparator regime*). The occupancy-weighted form is the natural object for trajectory-  
 1014 level base learners; the per-state KL/TV identity is preserved underneath. Information-directed  
 1015 sampling<sup>[20]</sup> requires a different analysis here because  $H(\pi^*) = 0$  collapses its information ratio at  
 1016 the deterministic- $\pi^*$  corner; we leave the IDS analysis as future work.

1017 **(b) Multi-factor forgetting schedule satisfying (A2)** Choose per-element discount rates  $\{\lambda_{ij}\}$   
 1018 such that the bottleneck  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min_{(i,j)} \nu_{ij} \lambda_{ij} (1 - \lambda_{ij})$  exceeds  $\rho_{\Sigma}/R_{\Sigma}$ . When  $\rho_{\Sigma}$  is unknown,  
 1019 the<sup>[2]</sup> black-box reduction gives an adaptive choice (applied per element if heterogeneous); when  $\rho_{\Sigma}$   
 1020 is known via a variation-budget<sup>[1]</sup> or change-point detection<sup>[37]</sup>, the choice is direct. As a fail-fast  
 1021 pre-check,  $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$  is necessary.

1022 **(c) Identifiability check satisfying (A3)** Test whether the loop data is in Regime A (full identifiability),  
 1023 Regime B (partial), or Regime C (none). In Regime A, no adjustment is needed. In Regime  
 1024 B, apply<sup>[17]</sup> (DOVI)-style confounding-bias adjustment. In Regime C, the bound is provable but not  
 1025 realizable; algorithm flags the regime as out-of-scope.

1026 **(d) Two-gap diagnostic readout satisfying (A4)** Compute  $\delta_{\text{sat}}$  (requires  $A_O$ , intractable in general;  
 1027 estimable via the policy-class supremum over recent rounds) and  $\delta_{\text{regret}}$  (requires  $V_O$  at current policy;  
 1028 tractable in simulation). Route to corrective action class.

1029 The algorithm is sketch-grade; full instantiation and empirical evaluation are deferred to a follow-up  
 1030 paper.

## E Pinsker vs. Point-Mass Identity Numerical Comparison

We compare  $V_{\max}(1 - e^{-D_{\text{KL}}})$  (point-mass identity bound, exact under deterministic  $\pi^*$  per Lemma 5.1 and the two-sided regret theorem) with  $V_{\max}\sqrt{D_{\text{KL}}/2}$  (Pinsker bound, also valid under deterministic  $\pi^*$  but loose),  $V_{\max}\sqrt{1 - e^{-D_{\text{KL}}}}$  (Bretagnolle–Huber inequality, the general envelope below which the identity sits at this corner), and the trivial envelope  $V_{\max}$ . Set  $V_{\max} = 1$  for normalization.

Table 2: Identity vs. Pinsker vs. BH at  $V_{\max} = 1$ .

| $D_{\text{KL}}$ | $1 - e^{-D_{\text{KL}}}$ (identity) | $\sqrt{1 - e^{-D_{\text{KL}}}}$ (BH) | $\sqrt{D_{\text{KL}}/2}$ (Pinsker) | ratio |
|-----------------|-------------------------------------|--------------------------------------|------------------------------------|-------|
| 0.01            | 0.00995                             | 0.0998                               | 0.0707                             | 7.10× |
| 0.1             | 0.0952                              | 0.308                                | 0.224                              | 2.35× |
| 0.5             | 0.393                               | 0.627                                | 0.500                              | 1.27× |
| 1.0             | 0.632                               | 0.795                                | 0.707                              | 1.12× |
| 2.0             | 0.865                               | 0.930                                | 1.000                              | 1.16× |
| 4.0             | 0.982                               | 0.991                                | 1.414                              | 1.02× |
| 10.0            | 0.99995                             | 0.99998                              | 2.236                              | 1.00× |

The point-mass identity bound is uniformly tighter than Pinsker, by a factor of  $7\times$  at  $D_{\text{KL}} = 0.01$  and converging to 1 as  $D_{\text{KL}}$  grows large (where both saturate at  $V_{\max}$ ). It is also uniformly tighter than the BH inequality  $\sqrt{1 - e^{-D_{\text{KL}}}}$  — the relation  $x < \sqrt{x}$  on  $(0, 1)$  — although both saturate together at  $V_{\max}$  for large  $D_{\text{KL}}$ . Pinsker, by contrast, hits the trivial envelope  $V_{\max} = 1$  at  $D_{\text{KL}} = 2$  and exceeds it beyond — vacuous from  $D_{\text{KL}} = 4$  onward (where the unclipped form returns 1.414), fully vacuous by  $D_{\text{KL}} = 10$ .

The matching lower bound  $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$  has the same shape, scaled by  $\Delta_{\min}/V_{\max}$ . In the extremal value landscape ( $\Delta_{\min} = V_{\max}$ ), the upper and lower bounds coincide and the regret is *identified exactly* by the KL coordinate. For typical landscapes, regret and KL are Lipschitz-equivalent with constants  $\Delta_{\min}/V_{\max}$  and 1.

## F Prior-Art Retrieval Summary

We conducted a structured prior-art retrieval (via Undermind) with the goal of identifying any framework composing all four target elements: (1) goal-feasibility-vs-policy-quality two-gap diagnostic; (2) an exact reverse-KL/TV regret identity (point-mass) under deterministic- $\pi^*$ , or — more loosely — Bretagnolle–Huber-style use in regret analysis; (3) strategic-tempo / forgetting prerequisite tying convergence to rate of useful policy revision; (4) closed-loop interventional access making regret bounds learnable from on-policy data.

**Result: 63 papers retrieved, abstract-level coverage estimated 75%, no direct anticipation in the retrieved set** The landscape splits into four largely separate strands corresponding to our four components; we did not find a published framework composing all four in this retrieval. Retrieval queries used the structured prior-art tool Undermind; the four-strand cite-and-distinguish in Section 2 surfaces the closest neighbor in each strand.

**Strongest negative signal (in the retrieved corpus): the Bretagnolle–Huber inequality is absent as an upper-bound coordinate for regret analysis in the retrieved RL / non-stationary literature** The information-theoretic regret line —<sup>[19–24]</sup> — uses entropy / mutual information / information ratio / Pinsker / Hellinger; we did not find the BH inequality deployed *as an upper-bound regret coordinate*. (BH is well-known in bandit *lower-bound* proofs via Le Cam’s method and Tsybakov’s framework<sup>[33]</sup>, including in standard RL textbook treatments<sup>[34]</sup>; that use is not contested here.) The *exact point-mass identity at the deterministic- $\pi$  corner\** (Lemma 5.1) — which strictly improves both Pinsker and the BH envelope at this corner — was likewise not found in the retrieved corpus, in either upper-bound or lower-bound use. We frame this as a literature-search observation against an explicitly bounded retrieval (63 papers, ~75% abstract-level coverage), not a formal absence claim against all published RL work; the claim is best read as “to our knowledge / in the retrieved set” rather than “absolutely novel.”



1070 **Closest neighbors per strand:**<sup>[1]</sup> (Strand 1 — variation-budget dynamic regret);<sup>[5]</sup> (Strand 2 —  
1071 closest two-term decomposition, different axis);<sup>[8]</sup> (Strand 3 — closest tempo result, different form);<sup>[14]</sup>  
1072 (Strand 4 — closest causal-RL, stationary only).

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1079 four-component composition (two-gap diagnostic, point-mass reverse-KL/TV identity, multi-  
1080 factor strategic tempo with forgetting prerequisite, closed-loop interventional access) and  
1081 the cumulative dynamic regret rate. Each component is developed in the main-result and  
1082 mechanism sections and proved in full in the appendices.

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